



NG

RP // INTERVIEW SERIES

JUNIOR → SENIOR

VOL 01 · FRONTEND INTERVIEW

Angular Interview Roadmap.

Junior → Senior

For Angular developers preparing for product-company interviews — concepts, patterns, and the answers that get you hired.

Signals

Change Detection

RxJS

NgRx

SSR

Performance

Testing

130+

PAGES

90+

QUESTIONS

8 MB

PDF SIZE

AUTHORED BY

Rutik Pimpale

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NAVIGATION

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TOPIC 1: ANGULAR BASICS

Q1: What is Angular? (IMPORTANT)

A: Angular is a **TypeScript-based frontend framework developed by Google** for building **single-page applications (SPAs)**. Unlike libraries, Angular is a **complete solution** — it comes with built-in support for **routing, forms, HTTP handling, dependency injection, and state management**, so we don't need to rely on third-party packages for basic functionality.

What makes Angular powerful is its **opinionated architecture** — it enforces a clean structure using **modules, components, and services**, which is very helpful when working in teams or on large-scale projects.

Real Example: In my project, we built a complete admin dashboard as a SPA using Angular. The entire application — user management, reports, settings — runs without a single page reload, and the **modular structure** made it easy for 4 developers to work on different features simultaneously.

Angular is the .

- *What is the difference between Angular and AngularJS?* — AngularJS (v1) was the old version built on **JavaScript** with **\$scope-based two-way binding**. Angular (v2+) is a **complete rewrite** using **TypeScript, component-based architecture**, and is significantly **faster and more maintainable**.
- *Why Angular over React?* — Angular is a **full-fledged framework** with routing, forms, DI, and HTTP built-in. React is a **UI library** — you need to add React Router, Redux, Axios, etc. separately. For large enterprise apps, Angular gives a **consistent structure out of the box**.

Q2: What is TypeScript and why does Angular use it? (IMPORTANT)

A: TypeScript is a **strongly-typed superset of JavaScript** developed by Microsoft. Angular chose TypeScript as its primary language because it provides **compile-time type checking, better IDE intelligence** (autocomplete, refactoring, error highlighting), and makes **large codebases much easier to maintain and debug**.

The biggest advantage I've experienced is that TypeScript catches bugs . For example, if I accidentally pass a string where a number is expected, my IDE immediately shows a red underline — I don't have to wait until the app runs and crashes.

Real Example: In my project, we had a user object with 15+ properties. With TypeScript **interfaces**, every developer knew exactly what properties were available, and the IDE auto-suggested them — this **reduced bugs by a significant margin** and made onboarding new developers faster.

TypeScript brings to Angular development.

- *Can we use Angular without TypeScript?* — Technically yes with plain JavaScript, but it's **strongly not recommended**. You lose type safety, decorator support, and the powerful tooling that makes Angular productive.

Q3: What is the difference between JIT and AOT compilation? (IMPORTANT)

A: **JIT (Just-in-Time)** compiles the Angular application **in the browser at runtime** — this is what happens during development when we run `ng serve`. **AOT (Ahead-of-Time)** compiles everything **during the build process itself** — this is used for production builds with `ng build`.

AOT is preferred for production because the browser receives **pre-compiled code**, which means **faster rendering**, **smaller bundle size** (Angular compiler is not shipped), and **template errors are caught at build time** rather than failing silently in production.

Real Example: When we switched from JIT to AOT in our project, the initial load time **dropped noticeably** because the browser didn't have to compile templates anymore — it just rendered the pre-compiled output directly.

AOT gives you .

- *Which one does `ng serve` use?* — JIT by default (for faster rebuilds during development).
- *Which one does `ng build` use?* — AOT by default since Angular 9+ (Ivy compiler).

Q4: What are the main building blocks of Angular?

A: Angular has a well-defined set of **building blocks** that work together to create a structured application. Each one has a **specific responsibility**, which keeps the codebase organized and maintainable.

- **Components** — the core UI building block, each controlling a section of the screen
- **Templates** — the HTML view attached to a component
- **Modules** — organize related code into cohesive blocks (though **standalone components** are replacing this in newer Angular)
- **Services** — hold **business logic, API calls, and shared data** — keep components clean
- **Directives** — add custom behavior to DOM elements (`*ngIf` , `*ngFor` , `ngClass`)
- **Pipes** — transform displayed data in templates (like formatting dates or currency)
- **Routing** — handles navigation between different views without page reload

Real Example: In any feature I build — say a "User Management" module — I create a **component** for UI, a **service** for API calls, use **pipes** to format dates, **directives** to conditionally show elements, and **routing** to navigate between list and detail views.

These building blocks enforce **separation of concerns**, which is key for **scalable architecture**.

- *Which building block do you use most?* — **Components and Services** — almost every feature involves creating a component for UI and a service for data/logic.

Q5: What is Angular CLI? Name some common commands.

A: Angular CLI is a **powerful command-line tool** that automates repetitive tasks like project setup, code generation, building, testing, and deployment. It enforces **best practices** and consistent project structure across teams.

I use it — from generating components to building production bundles. It saves a lot of manual configuration and boilerplate work.

Command	Purpose
<code>ng new app-name</code>	Create new project with full setup
<code>ng serve</code>	Run dev server with live reload
<code>ng g c component-name</code>	Generate component with files
<code>ng g s service-name</code>	Generate service
<code>ng g m module-name</code>	Generate module
<code>ng build</code>	Build optimized production bundle
<code>ng test</code>	Run unit tests with Karma

Real Example: When I need a new feature, I just run `ng g c features/user-list` and Angular CLI creates the component with its HTML, CSS, spec file, and even updates the module — all in one command.

Angular CLI is an for any Angular developer.

- *What does `ng g c --skip-tests` do?* — Creates the component **without** the `.spec.ts` test file — useful for quick prototyping.

Q6: What is the difference between var, let, and const?

A: These are the three ways to declare variables in JavaScript/TypeScript. Since Angular runs on TypeScript, understanding this is fundamental. The key differences are **scope**, **reassignment**, and **hoisting behavior**.

Feature	var	let	const
Scope	Function-scoped	Block-scoped <code>{ }</code>	Block-scoped <code>{ }</code>
Reassignment	Allowed	Allowed	Not allowed
Redeclaration	Allowed	Not allowed	Not allowed
Hoisting	Hoisted (initialized as <code>undefined</code>)	Hoisted but not initialized (TDZ)	Hoisted but not initialized (TDZ)
Use in Angular	Avoid	For values that change	Default choice

```
// var - function-scoped, can leak out of blocks
if (true) { var a = 10; }
console.log(a); // 10 - leaked out!

// let - block-scoped, safe
if (true) { let b = 20; }
console.log(b); // ReferenceError

// const - block-scoped + immutable reference
const user = { name: 'Rutik' };
user.name = 'Angular'; // OK - mutating properties is fine
user = {}; // Error - can't reassign the reference
```

Real Example: In my Angular project, I use **const** by default for everything — services injected via DI, arrays, config objects. I use **let** only for loop counters or values that genuinely need reassignment. I **never use var** — it causes scope leaks and bugs that are hard to trace.

- Does **const** make an object immutable? — No. **const** only prevents **reassigning the variable**. Object properties can still change. For true immutability, use **Object.freeze()** or **readonly** TypeScript modifiers.
- What is TDZ (Temporal Dead Zone)? — The period between entering a block and the **let** / **const** declaration — accessing the variable there throws **ReferenceError**. This helps catch bugs early.

TOPIC 2: COMPONENTS & DATA BINDING

Q1: What is a Component in Angular? (IMPORTANT)

A: A component is the **fundamental building block** of any Angular application. It controls a **specific part of the UI** and encapsulates its own **logic, template, and styling**. Every Angular app is essentially a **tree of components** — starting from the root `AppComponent`.

Each component has 3 parts:

- **TypeScript class** — holds the data and logic
- **HTML template** — defines what the user sees
- **CSS/SCSS** — scoped styling for that component only

We define it using the `@Component` decorator, which provides metadata like the **selector** (custom HTML tag), **template**, and **styles**.

```
@Component({
  selector: 'app-hello',
  templateUrl: './hello.component.html',
  styleUrls: ['./hello.component.css']
})
export class HelloComponent {
  message = 'Hello World!';
}
```

Real Example: In my project, the dashboard page was a single component composed of child components — `HeaderComponent`, `SidebarComponent`, `ChartComponent`, `TableComponent`. Each handled its own UI and logic independently, making the code **modular and easy to maintain**.

Components enable .

- *What is a selector?* — It's a **custom HTML tag** like `<app-hello>` that we use to embed this component anywhere in other templates.
- *What is the difference between `templateUrl` and `template`?* — `templateUrl` links to an **external HTML file**. `template` allows writing **inline HTML** directly inside the decorator — useful for very small templates.

Q2: What is Data Binding? What are its types? (IMPORTANT)

A: Data binding is the mechanism that creates a **live connection between the component's TypeScript class and its HTML template**. It's what makes Angular apps dynamic — when data changes, the view updates automatically. Angular supports **4 types of data binding**:

Type	Syntax	Direction	Use Case
Interpolation	<code>{{ value }}</code>	Component → View	Display dynamic text
Property Binding	<code>[property]="value"</code>	Component → View	Bind element properties
Event Binding	<code>(event)="handler()"</code>	View → Component	Handle user actions
Two-way Binding	<code>[(ngModel)]="value"</code>	Both ways	Sync form inputs

Real Example: In a search feature, I use `[(ngModel)]` on the search input so the variable updates as the user types, and I can **filter results in real-time** without any extra code for syncing.

Data binding is the — it eliminates manual DOM manipulation entirely.

- *What module is needed for ngModel?* — `FormsModule` must be imported in the module.
- *What is the difference between property binding and interpolation?* — Interpolation only works for **string values**. Property binding can bind **any type** — boolean, object, array. For example, `[disabled]="isLoading"` requires property binding because it's a boolean.

Q3: How do you pass data between components? (IMPORTANT)

A: This is one of the most common tasks in Angular. There are **3 main approaches**, and I choose based on the **relationship between components**:

The parent binds data to a child property using property binding.

```
// Child component
@Input() userName: string;

// Parent template
<app-child [userName]="parentName"></app-child>
```

The child emits events that the parent listens to.

```
// Child component
@Output() dataEmitter = new EventEmitter<string>();
sendData() {
  this.dataEmitter.emit('Hello from child!');
}

// Parent template
<app-child (dataEmitter)="receiveData($event)"></app-child>
```

3. Between unrelated components — Shared Service with BehaviorSubject When components have no parent-child relation, a shared service acts as the **communication bridge**.

Real Example: In my project, the **sidebar component** emits the selected menu item via a shared service, and the **main content area** subscribes to it and loads the corresponding view — both components are siblings with no direct relationship.

Choosing the right approach depends on .

- *Why use BehaviorSubject instead of Subject?* — BehaviorSubject **holds the last emitted value**, so even if a component subscribes **after** the data was sent, it still receives the most recent value immediately.
- *Can we use @Input() for deeply nested components?* — We can, but it creates **prop drilling** (passing data through multiple levels). A shared service is **cleaner and more scalable** for deep nesting.

Q4: What is @Component decorator?

A: `@Component()` is a **TypeScript decorator** that marks a class as an Angular component and provides the **metadata** Angular needs to create and render it. Without this decorator, Angular has no way of knowing that a class should behave as a component.

Key metadata properties:

- **selector** — the custom HTML tag name (e.g., `app-user`)
- **templateUrl** — path to the external HTML file
- **styleUrls** — array of CSS/SCSS files for scoped styling
- **providers** — component-level service instances
- **changeDetection** — strategy for detecting changes (Default or OnPush)
- **standalone** — marks as standalone component (Angular 15+)

Real Example: When I set `changeDetection: ChangeDetectionStrategy.OnPush` in the decorator, Angular only re-checks this component when its `@Input()` reference changes — this **significantly**

improved performance in our data-heavy table component.

The `@Component` decorator is what **transforms a plain TypeScript class into a fully functional Angular component**.

Q5: What is String Interpolation?

A: String interpolation is the simplest form of data binding in Angular. It uses **double curly braces** `{{ }}` to display component data directly inside the HTML template. It's a **one-way binding** — data flows from the TypeScript class to the view only.

```
<h1>Welcome, {{ username }}!</h1>
<p>Total: {{ price * quantity }}</p>
<p>Status: {{ isActive ? 'Active' : 'Inactive' }}</p>
```

We can use **simple expressions** inside interpolation — variables, arithmetic, ternary operators, and method calls. But **complex logic** like if-else blocks, assignments, or loops is not allowed.

Real Example: In my project's header component, I display `{{ currentUser.name }}` and `{{ notificationCount }}` using interpolation — it's the quickest way to show dynamic data in templates.

Interpolation is .

- *Can we call a function inside interpolation?* — Yes, like `{{ getTotal() }}`, but it's a **performance anti-pattern** because the function gets called on **every change detection cycle**. Better to use a **variable or pure pipe** instead.

Q6: What are Smart and Dumb Components?

A: This is a popular architectural pattern in Angular (borrowed from React) that **separates responsibilities** between components for better reusability, testability, and maintainability.

Aspect	Smart (Container) Components	Dumb (Presentational) Components
Purpose	Manage data and logic	Display data and emit events
Services	Inject services, call APIs	No services injected
Inputs/Outputs	Rarely used	Heavy use of <code>@Input()</code> and <code>@Output()</code>
Routing	Usually routed components	Typically children only
Reusability	Low (feature-specific)	High (reusable everywhere)
Change Detection	Default	<code>OnPush</code> (performance boost)

Real Example: In my project's user management feature, `UserListContainerComponent` (smart) calls `UserService.getUsers()`, handles pagination state, and passes data down. `UserCardComponent` (dumb) just receives a `user` input and emits `edit` / `delete` events. The same `UserCardComponent` is reused in the dashboard, profile page, and search results — because it doesn't know where its data comes from.

```
// Smart - owns data + logic
@Component({ selector: 'app-user-list-container', ... })
export class UserListContainerComponent {
  users$ = this.userService.getUsers();
  constructor(private userService: UserService) {}
  onDelete(id: string) { this.userService.delete(id); }
}

// Dumb - pure input/output
@Component({
  selector: 'app-user-card',
  changeDetection: ChangeDetectionStrategy.OnPush,
  template: `
    <div>{{ user.name }}</div>
    <button (click)="delete.emit(user.id)">Delete</button>
  `
})
export class UserCardComponent {
  @Input() user!: User;
  @Output() delete = new EventEmitter<string>();
}
```

- **Reusability** — dumb components work in any context
- **Testability** — dumb components test with simple inputs, no service mocks
- **Performance** — dumb components easily use `OnPush`
- **Maintainability** — clear boundaries of "who does what"

- *Can dumb components have their own state?* — Yes, **UI-only state** like "is dropdown open" is fine. But **business state** (user lists, filters, auth) belongs in smart components or services.

TOPIC 3: DIRECTIVES

Q1: What are Directives in Angular? What are the types? (IMPORTANT)

A: Directives are **special instructions that tell Angular how to modify or manipulate DOM elements**. They are one of the most powerful features of Angular, allowing us to add dynamic behavior to templates. There are **3 types**:

— Components are technically directives with their own template. Every component we create is a directive.

2. Structural Directives — These **change the DOM structure** by adding, removing, or replacing elements:

- `*ngIf` — conditionally add/remove an element
- `*ngFor` — iterate over a collection and render elements
- `*ngSwitch` — switch between multiple views based on a condition

3. Attribute Directives — These **change the appearance or behavior** of an existing element:

- `ngClass` — dynamically add/remove CSS classes
- `ngStyle` — dynamically apply inline styles

Real Example: In my project, I use `*ngIf` to show a **loading spinner** while the API is fetching data, and once data arrives, `*ngFor` renders the list. I use `ngClass` to highlight the **active row** in a table when the user clicks on it.

Directives let you without writing manual DOM manipulation code.

- *Why do structural directives have a `*` (asterisk)?* — The `*` is **syntactic sugar**. Behind the scenes, Angular converts it into an `<ng-template>` with the directive applied. It's just a shorthand for cleaner syntax.

Q2: What is the difference between `*ngIf` and `[hidden]`? (IMPORTANT)

A: Both hide elements from the user, but they work **completely differently under the hood**. `*ngIf` **physically removes or adds** the element from the DOM, while `[hidden]` just applies `display: none`

— the element **stays in the DOM** but becomes invisible.

Feature	<code>*ngIf</code>	<code>[hidden]</code>
How it works	Removes/adds element from DOM	Hides with CSS <code>display:none</code>
DOM impact	Element doesn't exist when false	Element always exists in DOM
Performance	Better for heavy components	Better for frequent toggling
Lifecycle hooks	Triggers <code>ngOnInit</code> / <code>ngOnDestroy</code> each time	No lifecycle impact

Real Example: For a **modal popup** with complex form logic inside, I use `*ngIf` — no point keeping it in the DOM when it's closed. But for a **dropdown menu** that the user toggles frequently, I use `[hidden]` to avoid re-creating it every time.

The choice depends on whether you want to **destroy and recreate** or just **visually toggle** the element.

- *When would `[hidden]` be a bad choice?* — When the hidden element has **active subscriptions, timers, or heavy computations** running — it wastes resources because the component is still alive in the DOM.

Q3: How does `*ngFor` work? What is `trackBy`? (IMPORTANT)

A: `*ngFor` is a **structural directive** that iterates over an array and renders a template block for each item. It also gives access to useful variables like `index`, `first`, `last`, `even`, and `odd`.

```
<div *ngFor="let user of users; let i = index">
  {{ i + 1 }}. {{ user.name }}
</div>
```

`trackBy` is a **performance optimization** that tells Angular how to **uniquely identify each item** in the list. Without it, Angular has no way to know which items changed — so it **destroys and recreates the entire DOM list** on every update. With `trackBy`, Angular only **updates the specific items that actually changed**.

```
trackById(index: number, user: any) {
  return user.id;
}
```

```
<div *ngFor="let user of users; trackBy: trackById">
```

Real Example: I had a real-time user list refreshing every 10 seconds from an API. Without `trackBy`, the entire table **flickered on each refresh**. After adding `trackBy: trackById`, only the changed rows updated — **smooth and flicker-free**.

`trackBy` is a **must-have for any list that updates dynamically** — it prevents unnecessary DOM manipulation.

- *What happens if you don't use trackBy with large lists?* — Angular destroys and recreates **all DOM elements** even if only one item changed — causes **poor performance, flickering, and loss of element state** (like input focus).

Q4: What is ngClass and ngStyle?

A: Both are **attribute directives** used for **dynamic styling**, but they serve different purposes. `ngClass` works with **CSS classes** while `ngStyle` works with **inline styles**.

ngClass — dynamically adds or removes CSS **classes** based on conditions:

```
<div [ngClass]="{ 'active': isActive, 'disabled': isDisabled, 'highlight': isSelected }">
```

ngStyle — dynamically applies **inline styles**:

```
<div [ngStyle]="{ 'color': isError ? 'red' : 'green', 'font-size': fontSize + 'px' }">
```

Real Example: In my project, I use `ngClass` to toggle an `active` class on the currently selected navigation item, and `ngStyle` to dynamically set the **width of a progress bar** based on a percentage value from the API.

When to use which: Use `ngClass` when you have **predefined CSS classes** ready. Use `ngStyle` when you need to **compute style values dynamically** at runtime (like colors from a theme API).

Q5: What is ng-template, ng-container, and ng-content?

A: These three are **Angular-specific elements** that don't render any actual DOM — they serve special structural purposes:

- `<ng-template>` — Defines a **lazy template block** that Angular **doesn't render by default**. It only renders when explicitly told to — via structural directives, `*ngIf` `else`, or `ngTemplateOutlet`.



```
<div *ngIf="isLoading; else loading">Data loaded!</div>
<ng-template #loading><p>Loading...</p></ng-template>
```

- `<ng-container>` — A **logical grouping element** that **adds zero extra DOM nodes**. Perfect when you need to apply a structural directive without adding a wrapper `div` that breaks your CSS layout.



```
<ng-container *ngIf="isLoggedIn">
  <p>Welcome!</p>
  <p>Your dashboard</p>
</ng-container>
```

- `<ng-content>` — Enables **content projection** (similar to React's `children`). It allows a parent component to **inject custom content** into a child component's template.



```
←!— Child template →
<div class="card">
  <ng-content></ng-content>
</div>

←!— Parent using child →
<app-card>
  <p>This content goes inside the card!</p>
</app-card>
```

Real Example: In my project, I built a **reusable card component** using `<ng-content>` — different pages inject their own content into the same card layout. I used `<ng-container>` to apply `*ngIf` on a group of elements without adding unnecessary `div` wrappers.

These three are .

- *What is multi-slot content projection?* — Using `<ng-content select=".header">` and `<ng-content select=".body">` to project content into **specific named slots** based on CSS selectors.

TOPIC 4: PIPES

Q1: What is a Pipe in Angular? (IMPORTANT)

A: A Pipe is a **template-level data transformer** — it takes a value, processes it, and returns the formatted output **without modifying the original data** in the component. We apply pipes using the **| (pipe) symbol** directly in HTML templates.

```
{{ birthday | date:'dd/MM/yyyy' }}  
{{ price | currency:'INR' }}  
{{ name | uppercase }}  
{{ longText | slice:0:50 }}
```

Common built-in pipes: `date`, `currency`, `uppercase`, `lowercase`, `titlecase`, `slice`, `json`, `async`, `percent`, `number`

Real Example: In my project, API dates come in ISO format like `2024-01-15T10:30:00Z`. Instead of writing formatting logic in the component, I simply use `{{ createdAt | date:'dd MMM yyyy' }}` in the template — it displays as `15 Jan 2024`. Clean and reusable.

Pipes keep **display logic in the template** where it belongs, making components **leaner and more focused on business logic**.

- *What is the difference between pure and impure pipes?* — **Pure pipes** execute only when the **input reference changes** — very efficient. **Impure pipes** execute on **every change detection cycle** — powerful but can impact performance on large datasets.

Q2: How do you create a Custom Pipe? (IMPORTANT)

A: Creating a custom pipe involves 3 simple steps: create a class that implements `PipeTransform`, add the `@Pipe` decorator with a **name**, and implement the `transform()` method with your formatting logic.

```
import { Pipe, PipeTransform } from '@angular/core';

@Pipe({ name: 'capitalize' })
export class CapitalizePipe implements PipeTransform {
  transform(value: string): string {
    if (!value) return '';
    return value.charAt(0).toUpperCase() + value.slice(1);
  }
}
```

```
{{ 'hello world' | capitalize }}

```

Real Example: I created a `timeAgo` custom pipe that converts timestamps into human-readable format — `"2 hours ago"`, `"3 days ago"`. I used it across the **notifications list, activity feed, and comment sections** — one pipe, reused everywhere.

Custom pipes are perfect for that you need across multiple templates.

- *How do you pass parameters to a custom pipe?* — Add extra arguments to the `transform()` method: `transform(value: string, limit: number)` and use as `{{ text | truncate:50 }}`.

Q3: What is the difference between Pure and Impure Pipes?

A: This is an important performance concept. **Pure pipes** (default) only re-execute when the **input value reference changes** — Angular caches the result and reuses it. **Impure pipes** re-execute on **every single change detection cycle**, regardless of whether the input changed or not.

Feature	Pure Pipe	Impure Pipe
Default	Yes (default behavior)	Must set <code>pure: false</code>
When it runs	Only when input reference changes	On every change detection
Performance	Highly efficient	Can be expensive
Use case	Simple transforms (format, uppercase)	Filtering/sorting arrays that mutate

```
@Pipe({ name: 'filter', pure: false })
```

Real Example: I once created a filter pipe for a table and kept it as pure. It wasn't updating when array items were modified. Switching to `pure: false` fixed it — but I quickly realized the **performance cost** on a 500-row table. So I refactored to use **immutable data patterns** with a pure pipe instead.

Rule of thumb: Always prefer **pure pipes** and use **immutable data** — only use impure pipes as a last resort.

Q4: What is the Async Pipe? (IMPORTANT)

A: The `async` pipe is one of the **most powerful built-in pipes** in Angular. It **subscribes to an Observable or Promise** directly in the template, renders the latest emitted value, and **automatically unsubscribes** when the component is destroyed. This means **zero manual subscription management and zero memory leak risk**.

```
// Component — no subscribe() needed!
users$ = this.http.get<User[]>('/api/users');
```

```
<!-- Template -->
<div *ngFor="let user of users$ | async">
  {{ user.name }}
</div>
```

- No manual `.subscribe()` and `.unsubscribe()` boilerplate
- **Prevents memory leaks automatically**
- Works perfectly with **OnPush change detection**
- Results in **cleaner, more declarative code**

Real Example: In my project, I switched from manual subscriptions to `async` pipe across most components — it reduced our code by roughly 30% and eliminated several memory leak issues we were debugging.

The `async` pipe is the .

- *When would you NOT use `async` pipe?* — When I need to **manipulate or combine data** before displaying, or when the same observable is used **multiple times** in the template (creates multiple subscriptions). For the latter, I use `*ngIf` with `as` to solve it:

```
<ng-container *ngIf="users$ | async as users">
  <p>Total: {{ users.length }}</p>
  <div *ngFor="let user of users">{{ user.name }}</div>
</ng-container>
```

TOPIC 5: SERVICES & DEPENDENCY INJECTION

Q1: What is a Service in Angular? (IMPORTANT)

A: A service is a **reusable TypeScript class** that holds **business logic, API communication, or shared data** — basically anything that's not directly related to the UI. The idea is to keep components **thin and focused only on presentation**, while services handle the heavy lifting behind the scenes.

```
@Injectable({ providedIn: 'root' })
export class UserService {
  constructor(private http: HttpClient) {}

  getUsers(): Observable<User[]> {
    return this.http.get<User[]>('/api/users');
  }
}
```

Real Example: In my project, I have clearly separated services — **AuthService** handles login/logout/token management, **UserService** handles user CRUD operations, **NotificationService** manages toast messages. Each service has a **single responsibility**, making them easy to test and maintain.

Services enforce the **single responsibility principle** and are the backbone of **clean Angular architecture**.

- *Can a service inject another service?* — Absolutely! As long as both have **@Injectable()**. For example, my **UserService** injects **HttpClient** for API calls and **LoggerService** for tracking errors — this is a very common pattern.

Q2: What is Dependency Injection (DI)? (IMPORTANT)

A: Dependency Injection is a **design pattern** where Angular's DI framework **automatically creates and provides** the dependencies a class needs, instead of the class creating them manually. This makes our code **loosely coupled, highly testable, and easily maintainable**.

```
// Angular injects UserService automatically – no 'new' keyword needed
constructor(private userService: UserService) {}
```

I never write `new UserService()` anywhere in my code. Angular's **injector** creates the instance, manages its lifecycle, and provides the **same singleton instance** wherever it's needed. During testing, I can easily **swap the real service with a mock** — that's the power of DI.

Real Example: In my project, by using DI, I can inject a `MockUserService` during unit tests that returns fake data — no actual API calls, tests run **fast and reliably**.

DI is what makes Angular apps .

- *What is an Injector?* — The injector is the **engine that creates and manages service instances**. Angular has a **hierarchical injector system** — root injector (app-wide), module injectors, and component injectors — each level can have its own instance.

Q3: What is @Injectable()? What does providedIn: 'root' mean? (IMPORTANT)

A: `@Injectable()` is a decorator that tells Angular this class **can participate in the dependency injection system** — meaning it can be injected into other classes, and it can also have dependencies injected into itself (like `HttpClient`).

`providedIn: 'root'` is the recommended way to register a service. It means:

- The service is a **singleton** — only one instance exists for the entire app
- It's **tree-shakable** — if no component uses it, Angular removes it from the production bundle automatically
- **No manual registration** needed in any `providers[]` array

Value	Meaning
'root'	Single instance for the entire app (most common)
'any'	New instance for every lazy-loaded module
SomeModule	Scoped to a specific module

Real Example: My `AuthService` uses `providedIn: 'root'` because the same login state must be accessible everywhere in the app. But for a `FormHelperService` that only a specific feature module needs, I provide it at the **component level** to get a fresh instance per component.

`providedIn: 'root'` is the **modern, recommended way** to register Angular services.

- *What happens if I provide a service in a component's providers array?* — A **new instance** is created specifically for that component and its children. Each component tree gets its own isolated copy of the service.

Q4: What is the difference between `providedIn: 'root'` and adding to providers array?

A: Both register a service for dependency injection, but they differ in **scope, tree-shaking, and recommended usage**:

Feature	<code>providedIn: 'root'</code>	<code>providers: []</code> in module
Singleton	Yes, app-wide automatically	Yes, within that module's scope
Tree-shakable	Yes — unused services removed from bundle	No — always included
Registration	Automatic (in the service itself)	Manual (in module/component)
Recommended	Yes — Angular's official recommendation	Only for special scoping needs

Real Example: In my project, all shared services like `AuthService`, `ApiService`, `NotificationService` use `providedIn: 'root'`. But for a `FormValidationService` that needs a **separate instance per feature module**, I register it in the module's `providers[]` array.

Best Practice: Default to `providedIn: 'root'`. Only use `providers[]` when you **intentionally need multiple instances** or module-scoped services.

Q5: How do you share data between unrelated components using a Service?

A: The cleanest approach is using a **shared service with BehaviorSubject**. The service acts as a **centralized communication hub** — one component pushes data into it, and any other component can subscribe to receive updates in real-time.

```
@Injectable({ providedIn: 'root' })
export class SharedDataService {
  private messageSource = new BehaviorSubject<string>('default');
  currentMessage$ = this.messageSource.asObservable();

  updateMessage(message: string) {
    this.messageSource.next(message);
  }
}
```

```
// Component A — sends data
this.sharedService.updateMessage('Hello from A!');

// Component B — receives data
this.sharedService.currentMessage$.subscribe(msg => {
  this.message = msg;
});
```

Why BehaviorSubject over Subject? — BehaviorSubject **retains the last emitted value**. So even if Component B subscribes **5 seconds after** Component A sent the data, B still gets the latest value immediately. With plain Subject, it would have missed it.

Real Example: In my project, the **sidebar navigation** and **breadcrumb component** are siblings — when the user selects a menu item in the sidebar, the breadcrumb updates automatically through a shared **NavigationService** using BehaviorSubject.

This pattern is the and works perfectly for most applications.

- *Why not just use @Input/@Output?* — Those only work for **parent-child** relationships. For **sibling or completely unrelated components**, a shared service is the cleanest and most scalable approach.

Q6: How does Angular's Dependency Injection Hierarchy work?

A: Angular's DI system is **hierarchical** — it mirrors the component tree. When a component requests a dependency, Angular walks **up the injector tree** until it finds a provider. This lets us **scope services at different levels** and even have **multiple instances** of the same service if needed.

1. **Platform Injector** — shared across all Angular apps on the page (rare case)
 2. **Root Injector** — created at app bootstrap; holds `providedIn: 'root'` services (singletons for the whole app)
 3. **Module Injectors** — for lazy-loaded modules, each gets its own injector
 4. **Component Injectors** — every component can declare its own `providers: []` — creates a **new instance** scoped to that component and its children
- Component asks for a service → Angular checks the **component's own injector** first
 - Not found? → moves up to the **parent component**
 - Keeps bubbling up → finally reaches **root injector**
 - If still not found → throws `NullInjectorError`

```
// Root-level singleton - ONE instance for entire app
@Injectable({ providedIn: 'root' })
export class AuthService { }

// Component-level - NEW instance per component tree
@Component({
  selector: 'app-wizard',
  providers: [WizardStateService] // fresh instance here
})
export class WizardComponent { }
```

Real Example: In my project, `AuthService` is `providedIn: 'root'` — one shared instance across the whole app. But `FormWizardService` is provided at the **component level** — each wizard instance gets its own isolated state, so opening two wizards in different tabs doesn't cause them to share data accidentally.

- **Singleton services** for cross-cutting concerns (auth, logging, API)
- **Scoped services** for per-feature or per-component state
- **Mockable for testing** — override a provider at the test bed level
- **Tree-shakable** — `providedIn: 'root'` services are removed from the bundle if unused
- *What happens if two injectors provide the same service?* — The **closest one wins**. A component-level provider overrides the root provider for that component and its children.

- What is `@Self()` and `@SkipSelf()`? — Decorators that control DI lookup. `@Self()` forces Angular to only look in the component's own injector. `@SkipSelf()` skips it and starts from the parent.

TOPIC 6: ROUTING

Q1: What is Routing in Angular? (IMPORTANT)

A: Routing in Angular enables **seamless navigation between different views/components** without triggering a full page reload — this is what makes it a true **Single Page Application**. We define a mapping between **URL paths and components**, and Angular's `RouterModule` handles the rest.

```
const routes: Routes = [  
  { path: '', redirectTo: '/home', pathMatch: 'full' },  
  { path: 'home', component: HomeComponent },  
  { path: 'users', component: UserListComponent },  
  { path: 'users/:id', component: UserDetailComponent },  
  { path: '**', component: NotFoundComponent }  
];
```

- `<router-outlet>` — the placeholder in HTML where the routed component gets rendered
- `routerLink` — used for declarative navigation in templates
- `Router.navigate()` — used for programmatic navigation in TypeScript
- `**` (**wildcard**) — catches all unmatched URLs and shows a custom 404 page

Real Example: In my project, I set up routing so `/dashboard` loads the `DashboardComponent`, `/users/:id` loads user details with the ID from the URL, and `**` shows a custom "Page Not Found" component. The navigation feels **instant** because Angular only swaps the component inside `<router-outlet>`.

Routing is what transforms a regular Angular app into a .

- What is `pathMatch: 'full'`? — It means the **entire URL path** must match exactly, not just the prefix. Used mainly with empty-path redirects to avoid catching all routes.

Q2: What is Lazy Loading? How do you implement it? (IMPORTANT)

A: Lazy loading is a **performance optimization technique** where feature modules are **loaded on demand** — only when the user actually navigates to that route — instead of being bundled into the initial app load. This **significantly reduces the initial bundle size** and makes the first page load much faster.

```
const routes: Routes = [
  {
    path: 'admin',
    loadChildren: () => import('./admin/admin.module')
      .then(m => m.AdminModule)
  }
];
```

```
{
  path: 'admin',
  loadChildren: () => import('./admin/admin.component')
    .then(c => c.AdminComponent)
}
```

Real Example: In my project, the Admin module had charts, reports, and user management — heavy code that only admins needed. After lazy loading it, the **main app bundle reduced by around 30%**, and regular users never download the admin code at all. This was one of the biggest performance wins we achieved.

Lazy loading is a for any production Angular application.

- *How do you verify lazy loading is working?* — Open the **Network tab** in browser DevTools. When you navigate to the lazy-loaded route, you'll see a **separate chunk file** being downloaded — that confirms it's working.
- *What is preloading strategy?* — After the initial load completes, Angular can **preload lazy modules in the background** using `PreloadAllModules` strategy — so they're ready instantly when the user navigates.

Q3: What are Route Guards? (IMPORTANT)

A: Route Guards are **security checkpoints** that control whether a user can **navigate to, leave, or load** a particular route. They act as gatekeepers — returning `true` to allow access or `false` (or a

redirect `UrlTree`) to deny it.

Guard	Purpose
<code>CanActivate</code>	Can the user access this route?
<code>CanDeactivate</code>	Can the user leave this route? (unsaved changes warning)
<code>CanActivateChild</code>	Can the user access child routes ?
<code>Resolve</code>	Fetch data before the route loads
<code>CanLoad</code>	Should the lazy module even be downloaded ?

```
export const authGuard: CanActivateFn = (route, state) => {
  const authService = inject(AuthService);
  if (authService.isLoggedIn()) {
    return true;
  }
  return inject(Router).createUrlTree(['/login']);
};
```

```
{ path: 'dashboard', component: DashboardComponent, canActivate: [authGuard] }
```

Real Example: In my project, I use `CanActivate` on every protected route to verify the user's **JWT token is valid** before allowing access. If the token is expired, the guard redirects to the login page. I also use `CanDeactivate` on form pages to **warn users about unsaved changes** before they navigate away.

Route Guards are in any real-world app.

- *What is the difference between `CanActivate` and `CanLoad`?* — `CanActivate` prevents **accessing** the route but the lazy module still gets downloaded. `CanLoad` prevents the **entire module from even being downloaded** — better for security since unauthorized users can't even see the code.

Q4: How do you pass data between routes?

A: Angular provides **3 distinct ways** to pass data during navigation, each suited for different scenarios:

```
// Route config
{ path: 'user/:id', component: UserComponent }

// Navigate
this.router.navigate(['/user', 123]);

// Read in target component
this.route.params.subscribe(params => {
  this.userId = params['id'];
});
```

```
// Navigate
this.router.navigate(['/users'], { queryParams: { page: 2, sort: 'name' } });
// URL: /users?page=2&sort=name

// Read
this.route.queryParams.subscribe(params => {
  this.page = params['page'];
});
```

```
this.router.navigate(['/user'], { state: { userData: this.user } });

// Read in target component
this.user = history.state.userData;
```

Real Example: In my project, I use **route params** for user detail pages (`/user/42`), **query params** for table filters and pagination (`/users?page=2&status=active`), and **router state** for passing the complete user object to an edit page without exposing it in the URL.

— params for required URL data, query params for optional filters, state for hidden complex data.

- What is the difference between **params** and **queryParams** ? — **params** are **part of the URL path** (`/user/123`) and are required. **queryParams** come **after the ?** (`/users?page=2`) and are optional. Params define the route, queryParams refine it.

Q5: What is a Resolver?

A: A Resolver is a special route guard that **pre-fetches data before the route activates**. The target component only loads **after the data is completely ready** — so the user never sees an empty screen or a flash of loading state.

```
export const userResolver: ResolveFn<User> = (route) => {  
  return inject(UserService).getUser(route.params['id']);  
};  
  
// Route config  
{ path: 'user/:id', component: UserComponent, resolve: { user: userResolver } }  
  
// Access in component — data is already available!  
constructor(private route: ActivatedRoute) {  
  this.user = this.route.snapshot.data['user'];  
}
```

In my project, I used a resolver for the user profile page — when you click on a user, the navigation waits until the API returns the user data, then renders the complete profile. No loading spinner needed.

Resolvers guarantee that **components always receive pre-loaded data**, creating a **smoother user experience**.

- *Downside of resolvers?* — Navigation **feels slower** because the user waits on a blank screen while data fetches. In many cases, it's actually better to load the component immediately and show a **skeleton loader** while the API call completes — feels more responsive.

TOPIC 7: FORMS (TEMPLATE-DRIVEN + REACTIVE)

Q1: What is the difference between Template-driven and Reactive Forms?

(IMPORTANT)

A: Angular provides **two approaches** for building forms, each designed for different complexity levels:

Feature	Template-Driven	Reactive
Defined in	HTML (template)	TypeScript (component)
Binding	<code>[(ngModel)]</code>	<code>FormControl</code> , <code>FormGroup</code>
Validation	HTML attributes (<code>required</code>)	<code>Validators.required</code> in TS
Module needed	<code>FormsModule</code>	<code>ReactiveFormsModule</code>
Best for	Simple forms (login, contact)	Complex forms (registration, dynamic)
Testing	Harder (DOM-dependent)	Easier (pure TypeScript)
Data flow	Two-way binding (implicit)	Unidirectional (explicit control)

My approach: For simple forms like login or contact, I use **Template-driven** because it's quicker to set up. For anything complex — **dynamic fields, conditional validation, multi-step wizards, FormArrays** — I always go with **Reactive Forms** because I get full programmatic control.

Real Example: In my project, the login page uses template-driven forms (just email + password). But the user registration form with dynamic address fields, conditional validation, and password matching uses **Reactive Forms** — much cleaner to manage in TypeScript.

Reactive Forms give you for complex scenarios.

- *Can you use both in the same project?* — Yes, absolutely. But not in the **same form**. Import both `FormsModule` and `ReactiveFormsModule` in your module.

Q2: How do you create a Reactive Form? (IMPORTANT)

A: I use Angular's `FormBuilder` service to create a `FormGroup` containing `FormControl` fields with `Validators`. This gives me **full control over form structure, validation, and data flow** — all in TypeScript.

```
import { FormBuilder, FormGroup, Validators } from '@angular/forms';

export class LoginComponent implements OnInit {
  loginForm!: FormGroup;

  constructor(private fb: FormBuilder) {}

  ngOnInit() {
    this.loginForm = this.fb.group({
      email: ['', [Validators.required, Validators.email]],
      password: ['', [Validators.required, Validators.minLength(6)]],
    });
  }

  onSubmit() {
    if (this.loginForm.valid) {
      console.log(this.loginForm.value);
    }
  }
}
```

```
<form [formGroup]="loginForm" (ngSubmit)="onSubmit()">
  <input formControlName="email" placeholder="Email">
  <div *ngIf="loginForm.get('email')?.errors?.['required']">
    Email is required
  </div>

  <input formControlName="password" type="password">
  <button type="submit" [disabled]="loginForm.invalid">Login</button>
</form>
```

Real Example: In my project, every form beyond basic login uses Reactive Forms. The registration form has **12+ fields with cross-field validation** (password match), conditional required fields, and dynamic sections — all managed cleanly through **FormBuilder**.

Reactive Forms are the .

- *How do you create a custom validator?* — Write a function that returns **null** (valid) or an error object (invalid):

```
function noSpaces(control: AbstractControl) {
  return control.value?.includes(' ') ? { noSpaces: true } : null;
}
```


Q3: What is FormGroup, FormControl, and FormArray?

A: These are the **three core building blocks** of Reactive Forms, each serving a different level of form structure:

- **FormControl** — represents a **single input field** (like email, name, or a checkbox). It tracks the value, validation state, and user interactions.
- **FormGroup** — a **collection of FormControls** grouped together (typically the entire form). Validation can also be applied at the group level.
- **FormArray** — a **dynamic array** of FormControls or FormGroups. You can **add or remove fields at runtime** — perfect for dynamic forms.

```
// FormArray example - adding multiple phone numbers dynamically
this.form = this.fb.group({
  name: [''],
  phones: this.fb.array([this.fb.control('')])
});

addPhone() {
  (this.form.get('phones') as FormArray).push(this.fb.control(''));
}
```

Real Example: I used **FormArray** in a form where users could add **multiple shipping addresses** — clicking "Add Address" dynamically creates a new address FormGroup with street, city, and pincode fields. The user can add or remove addresses freely.

FormArray is the that adapt to user needs at runtime.

Q4: How do you handle form validation? (IMPORTANT)

A: Angular provides a robust validation system with both **built-in validators** and support for **custom validators**. The key is combining proper validation rules with **user-friendly error messages** that only appear at the right time.

Built-in validators: `Validators.required`, `Validators.email`, `Validators.minLength()`, `Validators.maxLength()`, `Validators.pattern()`

```
<input formControlName="email">
<div *ngIf="loginForm.get('email')?.touched && loginForm.get('email')?.invalid">
  <span *ngIf="loginForm.get('email')?.errors?.['required']">Email is required</span>
  <span *ngIf="loginForm.get('email')?.errors?.['email']">Invalid email format</span>
</div>
```

- `touched` — user has visited and left the field
- `dirty` — user has actually changed the value
- `valid` / `invalid` — current validation status
- `errors` — object containing specific error details

Real Example: In my project, I show validation errors only when the field is **touched AND invalid** — this prevents showing errors before the user even interacts with the form. For the submit button, I use `[disabled]="form.invalid"` to prevent submission of incomplete forms.

Proper validation ensures .

- *What is cross-field validation?* — When validation depends on **multiple fields together** (e.g., password and confirm password must match). It's implemented at the **FormGroup level** as a group validator, not on individual FormControls.

Q5: What is ngModel and two-way binding?

A: `ngModel` enables **two-way data binding** between a form input and a component property. Changes in the input **instantly update the variable**, and changes to the variable **instantly update the input** — they stay perfectly in sync automatically.

```
// component.ts
username: string = '';
```

```
<!-- component.html -->
<input [(ngModel)]="username" placeholder="Enter name">
<p>Hello, {{ username }}!</p>
```

The `[( )]` syntax is called **"banana in a box"** — it's a combination of **property binding** `[]` (data flows in) and **event binding** `()` (data flows out). You must import `FormsModule` to use it.

Real Example: In my project's quick-search feature, I use `[(ngModel)]` on the search input — as the user types, the `searchTerm` variable updates instantly, and I use it to filter a local list in real-time

without any button click.

Two-way binding is .

Q6: How do Angular Forms work internally?

A: Internally, all Angular forms — whether template-driven or reactive — are built on the same **core abstractions** from `@angular/forms`. Understanding this helps you debug form issues and write custom form controls.

Class	Role
AbstractControl	Base class — every form control, group, and array extends it
FormControl	Represents a single input field — tracks value, validity, touched/dirty state
FormGroup	A collection of named controls — the form itself
FormArray	A dynamic array of controls — for repeating fields
ControlValueAccessor (CVA)	The bridge between DOM elements and FormControl — how Angular reads/writes values
NgForm / FormGroupDirective	Directives that connect templates to the model

1. You create a `FormControl` (either via `FormBuilder` in TS, or via `ngModel` in the template).
2. The **ControlValueAccessor** connects the DOM `<input>` to the `FormControl` — calling `writeValue()` when the model changes and notifying via `onChange()` when the user types.
3. Every keystroke → CVA emits the new value → `FormControl` updates its `value` → runs all attached **validators** → updates `status` (VALID/INVALID) → emits `valueChanges` and `statusChanges` observables.
4. Angular's **change detection** picks up the new state and re-renders the template (error messages, disabled states, etc.).

```
// Under the hood – every form field is an AbstractControl
form = new FormGroup({
  email: new FormControl('', [Validators.required, Validators.email])
});

// These Observables let you react to any change
this.form.valueChanges.subscribe(val => console.log(val));
this.form.statusChanges.subscribe(status => console.log(status));
```

Real Example: In my project, I built a **custom date-range picker** as a reusable form control by implementing `ControlValueAccessor`. Because I followed the same contract Angular's built-in inputs use, my custom component works seamlessly with Reactive Forms, validators, and `[(ngModel)]` — no extra wiring needed.

- You can build **custom form controls** that integrate perfectly with Angular's validation system
- You can tap into `valueChanges` for powerful reactive behavior (live search, auto-save, cross-field validation)
- Understanding this pipeline makes debugging form issues **much faster**
- *What are the 4 methods of `ControlValueAccessor`? — `writeValue()`, `registerOnChange()`, `registerOnTouched()`, and optional `setDisabledState()`.*
- *Which approach uses these classes more directly — template-driven or reactive? — **Reactive Forms** — you instantiate `FormGroup` / `FormControl` explicitly. Template-driven forms create them **implicitly** behind the scenes via `ngModel`.*

TOPIC 8: HTTP & API HANDLING

Q1: How do you make HTTP requests in Angular? (IMPORTANT)

A: Angular provides the `HttpClient` service from `@angular/common/http` for all HTTP communication. It returns **Observables** by default, which gives us powerful capabilities like **cancellation, retry, and operator chaining** through RxJS.

```
import { HttpClient } from '@angular/common/http';

@Injectable({ providedIn: 'root' })
export class UserService {
  private apiUrl = 'https://api.example.com';

  constructor(private http: HttpClient) {}

  getUsers(): Observable<User[]> {
    return this.http.get<User[]>(`${this.apiUrl}/users`);
  }

  createUser(user: User): Observable<User> {
    return this.http.post<User>(`${this.apiUrl}/users`, user);
  }

  updateUser(id: number, user: User): Observable<User> {
    return this.http.put<User>(`${this.apiUrl}/users/${id}`, user);
  }

  deleteUser(id: number): Observable<void> {
    return this.http.delete<void>(`${this.apiUrl}/users/${id}`);
  }
}
```

```
// In component
this.userService.getUsers().subscribe({
  next: (users) => this.users = users,
  error: (err) => console.error('Failed:', err)
});
```

Real Example: In my project, I follow a clean pattern — all HTTP calls live in **dedicated services**, components just call service methods and subscribe. This makes the code **testable** (I can mock the service) and **reusable** (multiple components use the same service).

HttpClient with Observables is the to handle API communication in Angular.

- *Why does HttpClient return Observable instead of Promise?* — Observables are **cancelable** (unsubscribe stops the request), support **retry logic**, can emit **multiple values**, and integrate seamlessly with **RxJS operators** for data transformation.

Q2: What are HTTP Interceptors? (IMPORTANT)

A: Interceptors are like **middleware for every HTTP request and response** in the app. They sit between the application and the server, allowing us to **modify requests or handle responses globally** — without changing individual service calls. Common uses include **attaching auth tokens, logging, global error handling, and showing loading spinners**.

```
import { HttpInterceptorFn } from '@angular/common/http';

export const authInterceptor: HttpInterceptorFn = (req, next) => {
  const token = localStorage.getItem('token');

  if (token) {
    const authReq = req.clone({
      setHeaders: { Authorization: `Bearer ${token}` }
    });
    return next(authReq);
  }

  return next(req);
};
```

```
provideHttpClient(withInterceptors([authInterceptor]))
```

Real Example: In my project, I had 2 interceptors working as a **pipeline**:

1. **Auth interceptor** — automatically attaches the JWT token to every outgoing request
2. **Error interceptor** — catches 401 responses globally and redirects to the login page

Without interceptors, I would have to add token logic in — interceptors saved hundreds of lines of duplicate code.

Interceptors are in any production Angular app.

- *Can you have multiple interceptors?* — Yes, they execute in the **exact order they are registered** — like a pipeline. The request passes through each interceptor sequentially.
- *Why do we clone the request?* — HTTP requests in Angular are **immutable objects**. We can't modify the original, so we **clone()** it with the changes we need.

Q3: How do you handle errors in HTTP calls? (IMPORTANT)

A: I handle errors at **multiple levels** for a robust error-handling strategy — service level for specific handling, and interceptor level for global catch-all behavior.

```
getUsers(): Observable<User[]> {  
  return this.http.get<User[]>('/api/users').pipe(  
    retry(2), // retry 2 times before failing  
    catchError(error => {  
      console.error('API Error:', error);  
      this.notificationService.showError('Failed to load users');  
      return of([]); // return empty array as fallback  
    })  
  );  
}
```

```
export const errorHandler: HttpInterceptorFn = (req, next) => {  
  return next(req).pipe(  
    catchError((error: HttpResponse) => {  
      if (error.status === 401) {  
        inject(Router).navigate(['/login']);  
      } else if (error.status === 500) {  
        inject(NotificationService).showError('Server error');  
      }  
      return throwError(() => error);  
    })  
  );  
};
```

Real Example: In my project, the error interceptor handles **401 (redirect to login)**, **403 (access denied toast)**, **500 (server error message)**, and **status 0 (network disconnected warning)**. Individual services handle specific cases like showing "No users found" when a search returns empty.

A ensures no error goes unnoticed while keeping the user informed at every step.

- What is the difference between `of()` and `throwError()` ? — `of()` returns a **fallback value** and the stream continues normally. `throwError()` **re-throws** the error so the subscriber's **error** callback catches it.

Q4: What is the difference between REST API and SOAP API?

A: REST and SOAP are two different **architectural approaches for building APIs**. In modern web development, REST has become the dominant choice due to its **simplicity and lightweight nature**.

Feature	REST API	SOAP API
Protocol	HTTP only	HTTP, SMTP, TCP
Data format	JSON (mainly), XML	XML only (strict)
Speed	Faster, lightweight	Slower, heavy overhead
Ease of use	Simple, flexible	Complex, strict standards
Stateless	Yes (each request is independent)	Can maintain state
Use case	Web apps, mobile apps, SPAs	Banking, enterprise, legacy systems

Real Example: In all my Angular projects, I've worked exclusively with **REST APIs** because they return **JSON** — which maps directly to TypeScript interfaces. Parsing and type-checking are effortless. SOAP with its XML envelope and strict schemas would be overkill for typical web applications.

REST APIs are the .

Q5: How do you implement caching in Angular?

A: Caching avoids **redundant API calls**, reduces server load, and makes the UI feel instant. There are several strategies depending on the use case — from simple in-memory caching to interceptor-level caching with TTL.

(simplest, for reference data)


```

@Injectables({ providedIn: 'root' })
export class CountryService {
  private countries$: Observable<Country[]>;

  getCountries(): Observable<Country[]> {
    if (!this.countries$) {
      this.countries$ = this.http.get<Country[]>('/api/countries').pipe(
        shareReplay(1) // cache the result, replay to any subscriber
      );
    }
    return this.countries$;
  }
}

```

First call hits the API. Every subsequent subscriber — no extra HTTP call.

```

@Injectables({ providedIn: 'root' })
export class CacheService {
  private cache = new Map<string, { data: any; expiry: number }>();

  get<T>(key: string, fetch: () => Observable<T>, ttl = 60000): Observable<T> {
    const cached = this.cache.get(key);
    if (cached && cached.expiry > Date.now()) {
      return of(cached.data);
    }
    return fetch().pipe(
      tap(data => this.cache.set(key, { data, expiry: Date.now() + ttl }))
    );
  }
}

```

(app-wide, transparent)

An interceptor caches GET responses by URL — subsequent identical requests skip the network entirely. Ideal for dashboards where the same lookups happen across many components.

Strategy 4: Browser storage — `localStorage` or `sessionStorage` for data that should survive page reloads (user preferences, feature flags).

Real Example: In my project, a `CountryService` and `CurrencyService` returned static lookup lists that changed once a week. Using `shareReplay(1)`, we went from **30+ API calls per session** to just **one** — a major load reduction on the backend and noticeably faster form rendering.

- Only cache **safe** data (GET requests, reference data) — never cache sensitive user-specific info on shared storage
- Always define a **clear invalidation strategy** — how the cache gets refreshed after edits

- Expose a `clearCache()` method for logout flows
- Difference between `shareReplay` and `publishReplay().refCount()`? — `shareReplay` is the modern, preferred operator — simpler API and works as expected in most cases.
- How do you invalidate the cache after a POST/PUT? — Reset the cached observable (`this.countries$ = undefined`) or call `cacheService.clear('countries')`.

Q6: How do you handle multiple API calls?

A: Handling multiple API calls is a very common interview scenario. RxJS gives us several operators — the right one depends on whether the calls are **parallel, sequential, or dependent** on each other.

(wait for all, combine results)

```
// Load dashboard data — 3 APIs in parallel
forkJoin({
  users: this.http.get<User[]>('/api/users'),
  orders: this.http.get<Order[]>('/api/orders'),
  stats: this.http.get<Stats>('/api/stats')
}).subscribe(({ users, orders, stats }) => {
  this.initDashboard(users, orders, stats);
});
```

Fires all 3 calls simultaneously. Emits , when all complete. If any fails, the whole thing fails.

```
// First get user, then get that user's orders
this.http.get<User>('/api/user/me').pipe(
  switchMap(user => this.http.get<Order[]>(`/api/orders?userId=${user.id}`))
).subscribe(orders => this.orders = orders);
```

When you want to show partial results as they stream in (e.g., notifications from multiple sources).

```
// Dashboard that re-renders whenever filter OR search changes
combineLatest([this.filter$, this.search$]).pipe(
  debounceTime(200),
  switchMap(([filter, search]) => this.api.query(filter, search))
).subscribe(results => this.results = results);
```

When each request must finish before the next starts (e.g., saving 10 items in order).

Real Example: In my project's analytics dashboard, `forkJoin` loads revenue, user stats, and recent orders in parallel — all 3 APIs fire together, and the UI shows a single spinner until all complete. This was 3× faster than the previous sequential implementation that loaded them one after another.

- **Independent, wait for all** → `forkJoin`
- **Each depends on the previous** → `switchMap` / `concatMap`
- **Need latest from multiple streams** → `combineLatest`
- **Fire and merge as they arrive** → `mergeMap` / `merge`
- *What if one API fails in `forkJoin`?* — The whole observable fails. To handle failures gracefully, wrap each inner call with `catchError(() => of(defaultValue))` so it "succeeds" with a fallback.
- *Difference between `forkJoin` and `combineLatest`?* — `forkJoin` emits **once, at the end**. `combineLatest` emits **every time any source emits** — continuous updates.

TOPIC 9: RXJS BASICS

Q1: What is RxJS and why is it used in Angular? (IMPORTANT)

A: RxJS (Reactive Extensions for JavaScript) is a **library for reactive programming** using **Observables**. Angular has RxJS deeply integrated — **HttpClient, Router events, Form value changes, ViewChild queries** — all return Observables. Understanding RxJS is essentially **mandatory for Angular development**.

- Can emit **multiple values** over time (not just one)
- **Cancelable** — `unsubscribe()` stops execution
- Has 100+ powerful **operators** for transforming, filtering, and combining data streams
- Perfect for **real-time scenarios** — WebSockets, live search, user input streams

Real Example: I use RxJS heavily for live search — `debounceTime(300)` waits for the user to stop typing, `distinctUntilChanged()` skips if the search term hasn't changed, and `switchMap` cancels the previous API call and fires a new one. This entire pipeline would take 30+ lines of imperative code — RxJS does it in **5 lines**.

RxJS is the and mastering it is what separates junior from mid-level developers.

Q2: What is the difference between Observable and Promise? (IMPORTANT)

A: Both handle asynchronous operations, but Observables are **significantly more powerful** and flexible than Promises:

Feature	Observable	Promise
Values	Multiple values over time	Only one value
Lazy	Yes — nothing happens until <code>subscribe()</code>	No — executes immediately on creation
Cancelable	Yes — <code>unsubscribe()</code> cancels	No — once started, can't stop
Operators	100+ operators (<code>map</code> , <code>filter</code> , <code>switchMap</code>)	Only <code>.then()</code> and <code>.catch()</code>
Built into Angular	Yes — core of HttpClient, Forms, Router	Supported but less common

Simple way to explain: A Promise is like **ordering a pizza** — you get one delivery and it's done. An Observable is like a **Netflix subscription** — you keep getting new content over time, and you can **cancel anytime** you want.

Real Example: In my project, I use Observables for API calls (HTTP), form value tracking, and real-time notifications. The ability to **chain operators** and **cancel ongoing requests** makes them far superior to Promises for Angular apps.

Observables are the .

Q3: What are commonly used RxJS operators? (IMPORTANT)

A: RxJS operators are **functions that transform, filter, or combine Observable streams**. Knowing the right operator for the right situation is crucial for writing efficient Angular code.

Operator	What it does	Real Use Case
<code>map</code>	Transform emitted values	Extract specific fields from API response
<code>filter</code>	Pass only values meeting a condition	Ignore empty search terms
<code>switchMap</code>	Cancel previous, switch to new Observable	Search-as-you-type (most asked!)
<code>mergeMap</code>	Handle multiple inner Observables in parallel	Parallel file uploads
<code>concatMap</code>	Handle Observables in strict sequence	Sequential form submissions
<code>debounceTime</code>	Wait N ms before emitting	Search input delay (avoid spam calls)
<code>distinctUntilChanged</code>	Skip consecutive duplicate values	Prevent duplicate API calls
<code>catchError</code>	Handle errors gracefully	Show fallback data on API failure
<code>takeUntil</code>	Auto-unsubscribe when signal fires	Memory leak prevention
<code>tap</code>	Side effects without changing data	Logging, debugging
<code>combineLatest</code>	Combine latest values from multiple streams	Dashboard needing data from 3 APIs

Most asked in interviews: `switchMap`, `debounceTime`, `catchError`, and `takeUntil` — know these with real examples and you'll impress the interviewer.

Q4: Explain `switchMap` with a real example. (IMPORTANT)

A: `switchMap` is the **most important RxJS operator for interviews**. It does one powerful thing — when a new value comes in, it **cancels the previous inner Observable** and switches to a new one. This makes it perfect for **search-as-you-type** scenarios.

Here's what happens:

- User types "An" → API call 1 starts
- User types "Ang" → API call 1 is **immediately canceled**, API call 2 starts
- User types "Angular" → API call 2 is **canceled**, API call 3 starts

Only the reaches the subscriber. No race conditions, no stale data.

```

this.searchControl.valueChanges.pipe(
  debounceTime(300),           // wait 300ms after user stops typing
  distinctUntilChanged(),      // skip if same search term
  switchMap(term =>            // cancel previous, make new API call
    this.http.get(`/api/search?q=${term}`)
  )
).subscribe(results => {
  this.searchResults = results;
});

```

Real Example: In my project's product search, before using `switchMap`, users sometimes saw **stale results** from an older, slower API call overwriting newer results. After implementing this pattern, the search became **perfectly reliable** — always showing results for the latest query.

`switchMap` is the **go-to operator for any scenario where only the latest request matters**.

- *When would you use `mergeMap` instead?* — When I **don't want to cancel** previous operations. Like saving multiple items to the server — each save should complete independently, regardless of new saves being triggered.

Q5: What is Subject and BehaviorSubject? (IMPORTANT)

A: Both are **special types of Observables** that can act as both **producer and consumer** — meaning they can emit values AND be subscribed to. The key difference is how they handle **late subscribers**.

Feature	Subject	BehaviorSubject
Initial value	No initial value	Requires an initial value
Late subscribers	Get nothing (miss all past values)	Get the last emitted value immediately
Use case	One-time events (button clicks, actions)	Sharing current state (logged-in user, settings)

```

// BehaviorSubject - always holds a current value
private userSubject = new BehaviorSubject<User | null>(null);
user$ = this.userSubject.asObservable();

login(user: User) {
  this.userSubject.next(user);
}

// Any component subscribing - even 10 seconds later - gets the current user

```

Real Example: I use `BehaviorSubject` in my `AuthService` to hold the **currently logged-in user**. The header component, sidebar, and profile page all subscribe to it. Even if the header component loads **after** login, it immediately gets the user data — no timing issues.

`BehaviorSubject` is the in Angular.

- *What about `ReplaySubject`?* — `ReplaySubject(N)` replays the **last N values** to new subscribers. Useful when you need a short history — like the last 3 notifications — not just the latest one.

Q6: When should you unsubscribe from Observables and why?

A: Unsubscribing prevents **memory leaks** — when a component is destroyed but its subscriptions keep running, they hold references in memory, may execute callbacks on a destroyed view, and can cause duplicate work or unexpected behavior.

Source	Why
<code>BehaviorSubject</code> / <code>Subject</code> streams in services	Long-lived; don't auto-complete
<code>interval</code> / <code>timer</code> (RxJS or <code>setInterval</code>)	Run forever until stopped
<code>WebSocket</code> / <code>SSE</code> / <code>EventSource</code>	Persistent connections
Form <code>valueChanges</code> / <code>statusChanges</code>	Live as long as the form lives
Router <code>events</code>	Continuous stream throughout the app
Manual <code>fromEvent</code> DOM listeners	Stay attached to the DOM

- **`HttpClient` calls** — they auto-complete after one emission
- `ActivatedRoute.params` / `queryParams` — Angular cleans them up
- **Observables consumed via `async` pipe** — pipe auto-unsubscribes on destroy
- **Anything wrapped with `take(1)` / `first()` / `takeUntilDestroyed()`** — completes itself

```
private destroy$ = new Subject<void>();

ngOnInit() {
  this.service.data$
    .pipe(takeUntil(this.destroy$))
    .subscribe(data => this.handle(data));
}

ngOnDestroy() {
  this.destroy$.next();
  this.destroy$.complete();
}
```

```
import { takeUntilDestroyed } from '@angular/core/rxjs-interop';

constructor() {
  this.service.data$
    .pipe(takeUntilDestroyed()) // auto-cleans on destroy
    .subscribe(...);
}
```

Real Example: In my project, a notification component had a `BehaviorSubject` subscription that was never unsubscribed. Every time the user navigated away and back, **a new subscription stacked on top of the old ones** — after 50 navigations, every incoming notification triggered the callback 50 times. Adding `takeUntil` immediately fixed it.

- **Memory leaks** — orphan subscriptions keep references alive forever
- **Duplicate side effects** — multiple stale subscribers reacting to one emission
- **Errors after destroy** — callbacks updating component state that no longer exists
- *Why does HTTP not need unsubscribing?* — Because `HttpClient` observables **complete after one emission**. Completed observables auto-clean their resources.
- *What's the safest single approach?* — Combine `async pipe` for templates + `takeUntilDestroyed()` for any imperative subscriptions. Together they handle 95% of real-world cases.

Q7: How do you optimize API calls with RxJS?

A: RxJS provides several powerful operators specifically designed to **reduce, debounce, cache, and combine** API calls — making apps faster, cheaper, and more reliable.


```
● ● ●  
  
this.search.valueChanges.pipe(  
  debounceTime(300) // ignore rapid input  
) .subscribe(...);
```

Reduces 10 keystrokes → 1 API call.

```
● ● ●  
  
.pipe(distinctUntilChanged()) // skip if same as previous emission
```

Prevents duplicate calls when the user retypes the same query.

```
● ● ●  
  
.pipe(switchMap(term ⇒ this.api.search(term)))
```

When a new request fires, the old one is — no stale results overwriting newer ones.

```
● ● ●  
  
this.config$ = this.http.get('/api/config').pipe(shareReplay(1));
```

First subscriber triggers the call. Every other subscriber gets the cached value — .

```
● ● ●  
  
forkJoin([this.api.users(), this.api.orders(), this.api.stats()])  
  .subscribe(([users, orders, stats]) ⇒ ...);
```

3 calls in parallel instead of sequential = ~3× faster.

```
● ● ●  
  
this.http.get('/api/data').pipe(retry({ count: 3, delay: 1000 })))
```

```
● ● ●  
  
.pipe(catchError(() ⇒ of([]))) // return empty list instead of failing
```

Real Example: A live-search feature in my project initially fired ~15 API calls per search session. After applying `debounceTime(300) + distinctUntilChanged() + switchMap`, it dropped to **1-2 calls per search** — an 85%+ reduction in backend load and a much smoother UX.

```

this.results$ = this.search.valueChanges.pipe(
  debounceTime(300),
  distinctUntilChanged(),
  filter(q => q.length >= 2),
  switchMap(q => this.api.search(q).pipe(
    catchError(() => of([]))
  )),
  shareReplay(1)
);

```

- What's the difference between `retry` and `repeat`? — `retry` resubscribes when the source **errors**. `repeat` resubscribes when the source **completes successfully** — useful for polling.
- How do you implement polling with RxJS? — `interval(5000).pipe(switchMap(() => this.api.getStatus()))` — fires every 5 seconds, cancels in-flight calls if a new tick comes.

TOPIC 10: LIFECYCLE HOOKS

Q1: What are Lifecycle Hooks? Name the important ones. (IMPORTANT)

A: Lifecycle hooks are **predefined methods** that Angular calls at **specific stages of a component's life** — from creation to rendering to destruction. They give us the ability to **run custom logic at exactly the right moment**.

Hook	When it runs	Common use
<code>constructor</code>	When the class is instantiated	Inject dependencies only
<code>ngOnChanges</code>	Before <code>ngOnInit</code> & whenever <code>@Input()</code> changes	React to parent data changes
<code>ngOnInit</code>	After first <code>ngOnChanges</code>	API calls, component initialization
<code>ngAfterViewInit</code>	After the view and child views are rendered	Access DOM via <code>@ViewChild</code>
<code>ngOnDestroy</code>	Just before the component is removed	Unsubscribe, cleanup resources

- `ngOnInit` — for initial API calls, setting up subscriptions, and initialization logic

- `ngOnDestroy` — for unsubscribing from Observables and cleaning up timers to prevent **memory leaks**

Real Example: In my project, every component that makes API calls follows this pattern — fetch data in `ngOnInit`, and clean up all subscriptions in `ngOnDestroy` using the `takeUntil` pattern.

Lifecycle hooks give you at every stage of its existence.

Q2: What is the difference between constructor and ngOnInit? (IMPORTANT)

A: This is one of the **most frequently asked Angular interview questions**. The constructor is a **TypeScript/JavaScript feature** — it runs when the class is instantiated. `ngOnInit` is an **Angular lifecycle hook** — it runs after Angular has fully set up the component, including its `@Input()` bindings.

Feature	constructor	ngOnInit
What is it	TypeScript class feature	Angular lifecycle hook
When it runs	When class instance is created	After <code>@Input()</code> bindings are ready
Use for	Only dependency injection	API calls, initialization logic
<code>@Input()</code> available?	No — not yet set	Yes — fully available

```
export class UserComponent implements OnInit {
  @Input() userId!: number; // NOT available in constructor

  constructor(private userService: UserService) {} // only inject

  ngOnInit() {
    // userId is available here - safe to use
    this.userService.getUser(this.userId).subscribe(user => {
      this.user = user;
    });
  }
}
```

Constructor = inject dependencies. ngOnInit = do the actual work. This keeps the code predictable and avoids timing issues.

- *Why not just do everything in the constructor?* — Because `@Input()` values are **not set yet** in the constructor. Angular sets them **after** construction but **before** `ngOnInit`. If you try to use `@Input()` data in the constructor, you'll get `undefined`.

Q3: What is ngOnDestroy? Why is it important? (IMPORTANT)

A: `ngOnDestroy` is the **cleanup hook** — it runs **just before Angular removes the component from the DOM**. Its primary purpose is to **free up resources** — especially **unsubscribing from Observables** to prevent **memory leaks** that can degrade app performance over time.

```
export class UserComponent implements OnInit, OnDestroy {
  private destroy$ = new Subject<void>();

  ngOnInit() {
    this.userService.getUsers()
      .pipe(takeUntil(this.destroy$))
      .subscribe(users => this.users = users);

    this.sharedService.data$
      .pipe(takeUntil(this.destroy$))
      .subscribe(data => this.data = data);
  }

  ngOnDestroy() {
    this.destroy$.next();
    this.destroy$.complete();
  }
}
```

- Clear `setInterval` / `setTimeout` references
- Close WebSocket connections
- Remove manual DOM event listeners
- Disconnect from third-party libraries

Real Example: In my project, we once had a **memory leak issue** — a component with a WebSocket subscription was being created and destroyed on route changes, but the subscription kept running in the background. After adding proper cleanup in `ngOnDestroy`, the memory leak disappeared and app performance stabilized.

`ngOnDestroy` is **non-negotiable for production apps** — skipping it leads to memory leaks, phantom subscriptions, and unpredictable behavior.

- *What happens if you don't unsubscribe?* — The subscription keeps running **even after the component is destroyed** — it becomes a **zombie subscription** that causes memory leaks, duplicate API calls, and unexpected side effects.

Q4: What is ngOnChanges? When does it fire?

A: `ngOnChanges` fires **every time an `@Input()` property changes** from the parent component. It provides a `SimpleChanges` object containing the **previous and current values**, plus whether it's the **first change**. This is very useful when you need to **react to input changes** and perform logic based on what changed.

```
ngOnChanges(changes: SimpleChanges) {  
  if (changes['userId'] && !changes['userId'].firstChange) {  
    // userId changed (not the initial load) - reload user data  
    this.loadUser(changes['userId'].currentValue);  
  }  
}
```

Important caveat: `ngOnChanges` only detects changes in **value/reference**, not mutations. If you mutate an object or push to an array, Angular won't detect it. You need to create a **new reference** — `[...array]` or `{...obj}` — for `ngOnChanges` to fire.

Real Example: In my project, I have a `UserDetailComponent` that receives `userId` as `@Input()`. When the parent changes the selected user, `ngOnChanges` detects it and **reloads the user data** — without this, the component would show stale data.

`ngOnChanges` is essential for components that need to **react dynamically to parent data changes**.

Q5: What is ngAfterViewInit?

A: `ngAfterViewInit` fires **after the component's view and all child views are fully initialized and rendered**. This is the earliest point where you can safely access **DOM elements** or **child components** via `@ViewChild`.

```
@ViewChild('chartCanvas') chartCanvas!: ElementRef;  
  
ngAfterViewInit() {  
  // Canvas element is now available in the DOM  
  this.initChart(this.chartCanvas.nativeElement);  
}
```

Real Example: In my project, I integrated a third-party charting library that needed a **reference to a canvas element**. I used `@ViewChild` to get the element and initialized the chart in `ngAfterViewInit`.

— trying to do it in `ngOnInit` would fail because the DOM isn't ready yet.

Important gotcha: If you modify component data inside `ngAfterViewInit`, you may get the dreaded `ExpressionChangedAfterItHasBeenChecked` error. The fix is to use `setTimeout()` or `ChangeDetectorRef.detectChanges()` to trigger a new change detection cycle.

`ngAfterViewInit` is the **go-to hook for any DOM manipulation or third-party library initialization**.

TOPIC 11: STATE MANAGEMENT (BASIC)

Q1: What is State Management? Why do we need it? (IMPORTANT)

A: State is the **data your application holds at any given moment** — the logged-in user, shopping cart items, applied filters, notification count, etc. State management is the strategy we use to **store, update, and share** this data predictably across all components that need it.

- In small apps, **services with BehaviorSubject** work perfectly fine
- In larger apps, when **multiple components** read and write the **same data**, things get messy — updates happen from different places, data goes out of sync, and bugs become hard to trace
- A proper state management approach gives you a **single source of truth** and a **predictable, traceable pattern** for all data changes

Real Example: In my project's e-commerce dashboard, the cart count is shown in the header, the cart details in the sidebar, and the total in the checkout page — all different components needing the **same cart state**. Using a centralized service with BehaviorSubject, any update to the cart **automatically reflects everywhere**.

State management ensures across the entire application.

- *Do you always need NgRx?* — No! For most mid-size apps, **services with BehaviorSubject** are more than enough. NgRx adds significant boilerplate and complexity — use it only when the app has **highly complex shared state** with many components reading and writing simultaneously.

Q2: How do you manage state using Services? (IMPORTANT)

A: I use a **service with BehaviorSubject** as a lightweight, centralized state store. The service holds the state privately, exposes it as an Observable for components to subscribe to, and provides **clean methods to update the state** — giving us a **mini Redux pattern** without any external library.

```
@Injectable({ providedIn: 'root' })
export class CartService {
  private cartItems = new BehaviorSubject<CartItem[]>([]);
  cartItems$ = this.cartItems.asObservable();

  addItem(item: CartItem) {
    const current = this.cartItems.value;
    this.cartItems.next([...current, item]);
  }

  removeItem(id: number) {
    const updated = this.cartItems.value.filter(i => i.id !== id);
    this.cartItems.next(updated);
  }

  getTotal(): number {
    return this.cartItems.value.reduce((sum, i) => sum + i.price, 0);
  }
}
```

Any component can subscribe to `cartItems$` and receive **automatic real-time updates** whenever the cart changes. The state is always **immutable** — we create new arrays instead of mutating.

Real Example: In my project, this exact pattern powers the cart, user preferences, and notification state. It's **simple, testable, and performant** — no external dependencies needed.

Service-based state management is the for most Angular applications.

Q3: What is NgRx? When would you use it?

A: NgRx is a **full-featured state management library** for Angular based on the **Redux pattern**. It provides a structured, predictable way to manage complex application state with strict rules about how state can change.

- **Store** — single source of truth holding all application state
- **Actions** — plain objects describing what happened ("Add Item", "Login Success")
- **Reducers** — pure functions that take current state + action and return new state
- **Selectors** — efficiently select and derive specific slices of state
- **Effects** — handle side effects like API calls, triggered by actions
- Large app with **complex shared state** across many features
- Multiple developers need a **strict, enforced pattern** for state changes
- You need **time-travel debugging** and **state change traceability**

When NOT to use: Small to medium apps — NgRx adds **significant boilerplate** (actions, reducers, effects for every feature). Service-based state management is simpler and sufficient for most cases.

Real Example: In my current projects, service-based state management handles everything well. But I understand the Redux pattern thoroughly and can adopt NgRx when the project demands it — especially in **large enterprise apps with 20+ developers** where enforced patterns prevent chaos.

NgRx is .

- *Have you used NgRx?* — "I understand the Redux pattern and NgRx concepts well. In my projects, service-based state management was sufficient. But I'm comfortable with the architecture and ready to use NgRx when the project requires it."

Q4: What are Angular Signals? (Angular 16+)

A: Signals are a **new reactive primitive** introduced in Angular 16 that provide a **simpler, synchronous way to manage reactive state**. A signal wraps a value and automatically notifies Angular when that value changes — no need for `subscribe()`, `async` pipe, or manual change detection.

```
import { signal, computed } from '@angular/core';

// Create a signal
count = signal(0);

// Computed signal (auto-updates when dependencies change)
doubleCount = computed(() => this.count() * 2);

// Update signal
increment() {
  this.count.update(val => val + 1);
}
```

```
<p>Count: {{ count() }}</p>
<p>Double: {{ doubleCount() }}</p>
<button (click)="increment()">+1</button>
```

- **Much simpler** than RxJS for component-level state
- **Better performance** — Angular knows exactly which parts of the template changed
- No `subscribe` / `unsubscribe` / memory leak concerns
- Key step toward **zoneless Angular** (removing Zone.js dependency)

Real Example: In newer parts of my project, I've started using Signals for **local component state** — like toggle flags, counters, and form status. For HTTP calls and complex async operations, I still use RxJS Observables. They complement each other well.

Signals represent the — definitely worth learning and adopting.

TOPIC 12: PERFORMANCE OPTIMIZATION

Q1: How do you improve Angular app performance? (IMPORTANT)

A: Performance optimization is something I actively think about in every project. Here are the **top strategies I follow and have implemented**:

1. Lazy Loading — Load feature modules only when the user navigates to them. This alone **reduced our initial bundle by 30%** in my project.

2. OnPush Change Detection — Skip unnecessary re-checks. Components only update when `@Input()` reference changes, events fire, or you manually trigger it.

— Prevent Angular from re-rendering entire lists. Only the changed items update in the DOM.

— Pre-compile templates at build time. Faster startup, smaller bundle, compile-time error checking.

5. Proper Unsubscription — Use `takeUntil` pattern or `async` pipe to prevent memory leaks from zombie subscriptions.

6. Avoid function calls in templates — `{{ calculate() }}` runs on **every change detection cycle**. Use computed values, variables, or **pure pipes** instead.

7. Pure Pipes over methods — Pipes with `pure: true` only re-execute when input reference changes — highly efficient.

8. Virtual Scrolling — For large lists (1000+ items), render only visible items using `@angular/cdk` ScrollingModule.

9. Bundle Analysis — Use `webpack-bundle-analyzer` to identify and remove heavy, unused dependencies.

10. Image Optimization — Use Angular's `NgOptimizedImage` directive for lazy loading and responsive images.

Real Example: In my project, combining lazy loading + OnPush + virtual scrolling brought the dashboard load time from **4 seconds to under 1.5 seconds** — a dramatic improvement that users immediately noticed.

Q2: What is OnPush Change Detection? How does it help? (IMPORTANT)

A: By default, Angular checks **every component in the entire tree** on every change detection cycle — even components whose data hasn't changed. **OnPush strategy** tells Angular to only re-check a component when one of these things happens:

- `@Input()` **reference** changes (not internal mutation)
- An **event** fires inside the component (click, keypress, etc.)
- You manually call `ChangeDetectorRef.markForCheck()`
- An **Observable bound with async pipe** emits a new value

```
@Component({
  changeDetection: ChangeDetectionStrategy.OnPush
})
export class UserListComponent {
  @Input() users!: User[];
}
```

Critical rule with OnPush — always use **immutable patterns**:

```
// WRONG - mutates array, OnPush won't detect it
this.users.push(newUser);

// CORRECT - creates new reference, OnPush detects it
this.users = [...this.users, newUser];
```

Real Example: In my project, after switching a heavy data table component (500+ rows) to OnPush, the change detection cycles **dropped by over 60%**. The table went from sluggish scrolling to **buttery smooth** — it was one of the biggest performance wins.

OnPush is a that every Angular developer should use for performance-critical components.

Q3: How do you handle memory leaks in Angular? (IMPORTANT)

A: Memory leaks in Angular typically happen when **subscriptions keep running** after a component is destroyed — the subscription becomes a "zombie" that consumes memory and can cause unexpected behavior. Here are the **3 approaches I use**:

```
private destroy$ = new Subject<void>();

ngOnInit() {
  this.service.data$.pipe(takeUntil(this.destroy$)).subscribe();
}

ngOnDestroy() {
  this.destroy$.next();
  this.destroy$.complete();
}
```

```
<div *ngFor="let user of users$ | async">{{ user.name }}</div>
```

Handles subscribe and unsubscribe — zero risk of leaks.

```
private sub!: Subscription;

ngOnInit() {
  this.sub = this.service.data$.subscribe();
}

ngOnDestroy() {
  this.sub.unsubscribe();
}
```

What you DON'T need to unsubscribe from: HTTP calls (they auto-complete after response) and **ActivatedRoute observables** (Angular handles them). But for long-lived Observables like **BehaviorSubject streams, interval timers, and WebSockets** — always clean up.

Real Example: We once had a component with 3 unsubscribed observables — every time the user navigated to and from that page, **3 new subscriptions stacked up**. After 50 navigations, we had 150 active subscriptions causing lag. Adding `takeUntil` fixed it completely.

Proper cleanup is .

Q4: What is Virtual Scrolling?

A: When you have a list of **thousands of items**, rendering all of them into the DOM at once makes the page **extremely slow and unresponsive**. Virtual scrolling solves this by **rendering only the items currently visible** in the viewport and **recycling DOM elements** as the user scrolls — so the DOM always contains just 20-30 elements, regardless of list size.

```
import { ScrollingModule } from '@angular/cdk/scrolling';
```

```
<!-- Template -->
<cdk-virtual-scroll-viewport itemSize="50" style="height: 400px;">
  <div *cdkVirtualFor="let item of items">
    {{ item.name }}
  </div>
</cdk-virtual-scroll-viewport>
```

Real Example: In my project, I had a transaction history page with **10,000+ records**. Without virtual scrolling, the page took **3+ seconds to render** and scrolling was laggy. After implementing **cdk-virtual-scroll-viewport**, it became **instant** — because only the ~20 visible rows exist in the DOM at any time. The difference was night and day.

Virtual scrolling is the without compromising user experience.

Q5: What is Tree Shaking?

A: Tree shaking is a **build-time optimization** that automatically **removes unused code** from the final production bundle. Angular's build tools (webpack/esbuild) analyze the **import graph** to determine which functions, classes, and modules are actually used — everything else gets "shaken off" like dead leaves from a tree.

How it works in practice: If I install **lodash** (a 70KB library) but only use **_.debounce**, tree shaking removes all other lodash functions and only includes **debounce** in the bundle — saving significant bundle size.

- Use **providedIn: 'root'** for services — they're **tree-shakable by default**
- Use **ES module imports** (**import { debounce } from 'lodash-es'**), not CommonJS **require**
- Avoid wildcard imports (**import * as _ from 'lodash'** — prevents tree shaking)

Real Example: In my project, I switched from **import * as moment from 'moment'** to the lighter **date-fns** library with individual imports — combined with tree shaking, it **reduced our bundle by 200KB**.

Tree shaking ensures your production bundle contains — nothing more.

Q6: What is Change Detection and how does it work internally?

A: Change Detection is Angular's mechanism to **sync the component's data model with the DOM**. Whenever something changes — a button click, an HTTP response, a timer — Angular walks the component tree, checks for changes, and updates the view where needed.

1. **Zone.js monkey-patches async APIs** — `setTimeout`, `addEventListener`, `Promise.then`, XHR, etc. Any async operation triggers Angular's zone, notifying it "something may have changed."
2. Angular then runs `ApplicationRef.tick()` — walks the **entire component tree** from root to leaves.
3. For each component, Angular compares current bound values against the previous snapshot.
4. If a binding changed → update the DOM.
5. In **dev mode**, Angular runs a **second pass** to catch `ExpressionChangedAfterItHasBeenCheckedError` — an expression that changes between the two passes indicates a bug.

Feature	Default	OnPush
Checks every CD cycle	Yes	Only on specific triggers
Triggers	Any async event anywhere	<code>@Input</code> reference change, event in component, async pipe emission, <code>markForCheck()</code>
Performance	Slower for big trees	Much faster

```

Async event (click / HTTP / timer)
  ↓
Zone.js detects it
  ↓
ApplicationRef.tick()
  ↓
Traverse component tree (top → bottom)
  ↓
For each component: compare bindings
  ↓
Update DOM where values changed

```

Real Example: In my project, a dashboard with 40+ components was re-checking every component on **every keystroke** in a search box. By switching performance-heavy widgets (charts, tables) to **OnPush**, we reduced change detection time per cycle from **~25ms to ~6ms** — dramatically smoother interaction.

Angular is moving away from Zone.js. With **signals** (v16+), change detection becomes **fine-grained** — Angular knows exactly which components depend on which data and only updates those. Zoneless Angular (v18+ experimental) eliminates Zone.js entirely, relying on signals.

- *What causes `ExpressionChangedAfterItHasBeenCheckedError`?* — Modifying a bound value **inside lifecycle hooks that run during/after change detection** (like `ngAfterViewInit`). Fix with `setTimeout()`, `markForCheck()`, or refactoring the logic.
- *What does `markForCheck()` do?* — Marks the component (and its ancestors) as "needs check" on the next change detection cycle. Essential with `OnPush` when updating from outside normal triggers.

Q7: What is Zone.js and what role does it play in Angular?

A: Zone.js is a library that **monkey-patches browser async APIs** — `setTimeout`, `setInterval`, `Promise`, `addEventListener`, XHR, etc. — to notify Angular when async work happens. Without Zone.js, Angular wouldn't know **when to run change detection**.

1. Creates an execution context called a **Zone** around the app
2. Wraps every async operation — so Angular gets a notification (`onMicrotaskEmpty`) when async work finishes
3. Triggers `ApplicationRef.tick()` → runs change detection → updates the DOM

Without Zone.js — you'd have to manually call `changeDetectorRef.detectChanges()` after every async operation. Zone.js is what makes Angular's "magic auto-updates" work.

```
Browser API (setTimeout)
  ↓ (Zone.js intercepts)
Runs callback
  ↓ (Zone.js notifies Angular)
Angular runs change detection
  ↓
DOM updates
```

- Adds ~100KB to the bundle
- Patches every async API — occasional compatibility issues with third-party libs
- Triggers full-tree change detection on **every** async event — inefficient for large apps

Angular is moving toward removing Zone.js entirely. The modern stack — **Signals + standalone components** — lets Angular do change detection **only for components that actually changed**.

```
// main.ts - opt into zoneless experimental mode
bootstrapApplication(AppComponent, {
  providers: [provideExperimentalZonelessChangeDetection()]
});
```

Real Example: In my project, third-party WebSocket libraries sometimes ran **outside the Angular zone** — so UI updates didn't reflect until the user clicked something. Fixed by wrapping callbacks in `NgZone.run()` to re-enter the zone. Alternatively, `runOutsideAngular()` is useful when you want to run heavy background work **without triggering change detection** on every tick.

- *When would you use `NgZone.runOutsideAngular()`?* — For CPU-heavy work like animations, chart rendering, or high-frequency events (scroll/mousemove) — you don't want change detection running 60+ times per second.
- *Can you fully remove Zone.js?* — Yes, in Angular 18+ with `provideExperimentalZonelessChangeDetection()` and a signal-based app. Still experimental but production-ready in many apps.

Q8: How do you reduce the bundle size of an Angular app?

A: Bundle size directly impacts **initial load time, Time-to-Interactive, and mobile performance**. Here's the full toolkit I use to keep bundles lean in production Angular apps.

— biggest win

```
{ path: 'admin', loadComponent: () => import('./admin/admin.component')
  .then(m => m.AdminComponent) }
```

Splits the bundle into chunks downloaded only when needed.

(Angular 17+) — load sections on demand

```
@defer (on viewport) {
  <app-heavy-chart />
}
```

Defers heavy components until they're actually needed.

— remove unused code

- Use `providedIn: 'root'` for services (tree-shakable)

- Use **ES module imports** (`import { debounce } from 'lodash-es'`) — not `import * as _ from 'lodash'`
- Keep `sideEffects: false` in library `package.json` where possible

Heavy	Lighter
moment.js (~70KB)	date-fns (tree-shakable) or <code>Intl.DateTimeFormat</code> (built-in)
lodash	Native JS methods or lodash-es individual imports
rxjs full	Pipeable operators only — already tree-shaken with RxJS 7+

```
ng build --configuration=production
```

Enables AOT, minification, uglification, dead-code elimination, and `buildOptimizer`.

— prevent regressions

```
// angular.json
"budgets": [
  { "type": "initial", "maximumWarning": "500kb", "maximumError": "1mb" }
]
```

Build fails if bundle grows beyond limits.

```
ng build --stats-json
npx webpack-bundle-analyzer dist/stats.json
# or for esbuild builds
npx source-map-explorer dist/**/*.js
```

Identifies which packages are bloating your bundle.

— enables more aggressive tree shaking of change detection code paths.

— reduces image payload with lazy loading and responsive sizing.

10. Avoid polyfills for browsers you don't support — check `polyfills.ts` and browserslist.

Real Example: In my project, combining **lazy loading + replacing moment with date-fns + removing unused lodash functions + `@defer` for charts** dropped our initial bundle from **2.1 MB to 620 KB** (gzipped). Lighthouse performance score jumped from 54 to 92.

- *What's the difference between `ng build` and `ng build --configuration=production`?* — Production enables AOT, optimization, hashing, no source maps. Development skips most of these

for faster rebuilds.

- *How do you identify the biggest offenders?* — Run bundle analyzer and look for large modules, duplicate dependencies, and accidentally-imported dev-only code.

TOPIC 13: ANGULAR INTERVIEW SCENARIO QUESTIONS

Q1: Tell me about a challenging problem you solved in Angular. (IMPORTANT)

A: "In my project, we had a **product search page** where users could search and filter from a list of 5000+ items. The problem was — every single keypress was triggering an **API call**, which caused the UI to lag, the server got hammered with unnecessary requests, and sometimes **older search results would overwrite newer ones** because of network timing.

I solved it using a combination of :

- `debounceTime(300)` — wait 300ms after the user stops typing before making the call
- `distinctUntilChanged()` — skip if the search term hasn't actually changed
- `switchMap` — automatically cancel the previous API call when a new one fires

This **reduced API calls by over 80%**, completely eliminated the stale results issue, and made the search feel **instant and responsive**. I also added **virtual scrolling** for the results list since it could return thousands of items — so the DOM only renders the visible rows.

The combination of transformed a sluggish feature into one of the smoothest parts of our application."

Q2: How do you structure a large Angular application? (IMPORTANT)

A: I follow a **modular, feature-based structure** that keeps the codebase **organized, scalable, and easy to navigate** — even when the team grows:

```
src/
├── app/
│   ├── core/           → Singleton services (AuthService, interceptors, guards)
│   ├── shared/         → Reusable components, pipes, directives
│   ├── features/       → Feature modules (each lazy loaded independently)
│   │   ├── dashboard/
│   │   ├── users/
│   │   └── settings/
│   ├── models/         → TypeScript interfaces and type definitions
│   └── app-routing.module.ts
```

- **Core module** — contains singleton services (AuthService, API interceptors, error handlers). Imported **only in AppModule** — never in feature modules.
- **Shared module** — contains reusable UI components (buttons, modals, cards), pipes, and directives. Imported by **any feature module** that needs them.
- **Feature modules** — each feature is **self-contained and lazy loaded**. A new developer can understand a feature by looking at one folder.
- **Separation of concerns** — components handle only UI/presentation, services handle all logic and API communication.

Real Example: In my project with 15+ features and 4 developers, this structure allowed each developer to work on their feature module **independently** without merge conflicts or tight coupling. Adding a new feature was as simple as creating a new folder under `features/`.

This architecture ensures .

Q3: How do you handle authentication in Angular? (IMPORTANT)

A: "I implement authentication using the **JWT (JSON Web Token) flow**, which is the most common approach for Angular SPAs:

1. Login Flow — User submits credentials → API validates and returns a **JWT token** → I store it in **localStorage** (or sessionStorage for stricter session control).

2. Auth Interceptor — A global HTTP interceptor that **automatically attaches** the token to every outgoing API request via `Authorization: Bearer <token>` header. No manual token handling in individual services.

3. Auth Guard (CanActivate) — Protects all private routes. Before allowing navigation, it checks if the user **has a valid, non-expired token**. If not → redirect to login page.

4. Token Expiry Handling — I decode the JWT to check expiry before critical API calls. If expired, either redirect to login or implement a **refresh token flow** to silently get a new token.

5. Auth Service with BehaviorSubject — Central service holding the current user state. Components subscribe to `currentUser$` to know the login status reactively — the header shows the username, guards check authentication, and the sidebar shows/hides menu items."

Real Example: In my project, this entire auth flow works seamlessly — login → interceptor adds token → guard protects routes → expired token triggers re-login. The user experience is smooth, and the code is **clean and centralized**.

- *Where do you store the token?* — `localStorage` for persistence across browser tabs. For better **XSS protection**, `httpOnly cookies` are ideal — but they require backend configuration and CSRF handling.

Q4: How do you handle errors globally in your app?

A: "I implement a **layered error-handling strategy** that covers errors at every level — from specific component failures to unexpected global exceptions:

— Show context-specific error messages for specific feature failures (e.g., "Failed to load user list — please retry").

2. Service level — Use RxJS `catchError` to handle and transform API errors — return fallback data or user-friendly error messages.

3. Global HTTP level (Error Interceptor) — A single interceptor catches **all HTTP errors** and handles them by status:

- **401 Unauthorized** → Clear token, redirect to login
- **403 Forbidden** → Show "Access Denied" notification
- **500 Server Error** → Show generic "Something went wrong" toast
- **0 (Network Error)** → Show "Please check your internet connection"

4. Unhandled Exceptions (Global ErrorHandler) — Angular's `ErrorHandler` catches any **uncaught JavaScript errors** across the app:

```
@Injectable()
export class GlobalErrorHandler implements ErrorHandler {
  handleError(error: any) {
    console.error('Global Error:', error);
    // Log to monitoring service (Sentry, LogRocket, etc.)
  }
}
```

Real Example: In my project, this layered approach ensures **no error goes unhandled** — HTTP errors show user-friendly messages, auth failures redirect cleanly, and any unexpected runtime error gets

logged to our monitoring dashboard for debugging.

Q5: Explain a situation where you optimized performance. (IMPORTANT)

A: "We had a **dashboard page** that was our main landing page after login, and it was loading painfully slow — taking **4+ seconds** to become interactive. I investigated and found **three major bottlenecks**:

Problem 1: Massive initial bundle All 8 feature modules were loading upfront, even though the user only needed the dashboard. **Solution:** Implemented **lazy loading** for all feature modules except the dashboard. Initial load time dropped from **4s to 1.8s** immediately.

Problem 2: Slow rendering of a large data table A table with 2000+ transaction rows was causing the browser to struggle with DOM rendering and scrolling. **Solution:** Added **virtual scrolling** from Angular CDK and `trackBy` in `*ngFor`. The table now renders only visible rows (~25 at a time) — scrolling became **instant and smooth**.

Problem 3: Excessive change detection Multiple dashboard widgets (charts, counters, tables) were all using Default change detection, causing Angular to re-check **everything** on every user interaction. **Solution:** Switched performance-critical components to **OnPush change detection** strategy. Change detection cycles **reduced by over 60%**.

Final Result: Page load time went from **4 seconds to under 1.5 seconds**, scrolling was smooth, and the dashboard felt **responsive and professional**. This was the most impactful performance improvement I delivered."

Q6: What is the difference between ViewChild and ContentChild?

A: Both are decorators for **querying elements** from the template, but they differ in **where** they look:

- **@ViewChild** — Accesses elements defined in the **component's own template** (its own HTML)
- **@ContentChild** — Accesses elements that are **projected into the component** from a parent via `<ng-content>`

```
// ViewChild - element in own template
@ViewChild('myInput') inputRef!: ElementRef;

// ContentChild - element projected from parent
@ContentChild('projectedItem') projectedItem!: ElementRef;
```

Real Example: In my project's reusable `CardComponent`, I use `@ContentChild` to access the projected header element and apply animations to it. For a chart component, I use `@ViewChild` to get

the canvas element defined in its own template.

Simple rule: Your own template → `@ViewChild` . Content coming from parent → `@ContentChild` .

Q7: What are Standalone Components? (Angular 15+)

A: Standalone components are a **major simplification** introduced in Angular 15 — they **don't require NgModule** to function. Instead, they declare their own dependencies directly in the `@Component` decorator. This reduces boilerplate and makes components **truly self-contained**.

```
@Component({
  selector: 'app-hello',
  standalone: true,
  imports: [CommonModule, RouterModule],
  template: `<h1>Hello, {{ name }}!</h1>`
})
export class HelloComponent {
  name = 'World';
}
```

- **No more NgModule boilerplate** — each component manages its own imports
- **Easier to understand** — everything a component needs is declared right there
- **Better tree-shaking** — unused components are removed more efficiently
- This is the **official future direction** of Angular — new projects should prefer standalone

Real Example: In newer features of my project, I've started using standalone components. The developer experience is **noticeably smoother** — no more jumping between component and module files to add imports. Each component is a **self-contained unit** ready to use anywhere.

Standalone components are the — Angular is moving away from NgModules.

Q8: What is the new Control Flow syntax in Angular 17+?

A: Angular 17 introduced **built-in control flow syntax** that replaces traditional structural directives with a **cleaner, more intuitive template syntax**. It's faster, requires no imports, and looks like modern template languages.

```

<!-- Old way with *ngIf -->
<div *ngIf="isLoggedIn; else loginTemplate">Welcome!</div>
<ng-template #loginTemplate>Please login</ng-template>

<!-- New way (Angular 17+) -->
@if (isLoggedIn) {
  <div>Welcome!</div>
} @else {
  <div>Please login</div>
}

<!-- Old *ngFor -->
<div *ngFor="let user of users; trackBy: trackById">{{ user.name }}</div>

<!-- New @for with built-in @empty -->
@for (user of users; track user.id) {
  <div>{{ user.name }}</div>
} @empty {
  <div>No users found</div>
}

<!-- New @switch -->
@switch (status) {
  @case ('active') { <span>Active</span> }
  @case ('inactive') { <span>Inactive</span> }
  @default { <span>Unknown</span> }
}

```

- **Cleaner, more readable** syntax — no more `ng-template` hacks for else blocks
- **Better performance** — optimized at the compiler level
- **Built-in `@empty` block** for `@for` — no need for separate `*ngIf` to check empty arrays
- **No CommonModule import needed** — these are built into the template compiler

Real Example: In my project, after migrating templates to the new syntax, code reviews became much easier — the control flow is **immediately readable** even for developers new to Angular.

The new control flow syntax is for Angular templates.

Q9: How do you improve Angular app scalability?

A: Scalability means the app handles **growing teams, growing codebases, and growing traffic** without collapsing under its own weight. I treat scalability as a combination of **architecture, performance, and developer ergonomics**.

1. Feature-based modular structure — each feature self-contained under `features/`, independently lazy-loaded, with its own data-access, UI, and routing. Teams can work in parallel without stepping on each other.

2. Enforce boundaries — in Nx workspaces, lint rules prevent features from importing each other directly. All sharing goes through `shared/` or `core/` libs. Prevents the "spaghetti imports" that kill large codebases.

3. Smart/Dumb component split — dumb components become **reusable design-system pieces**. Smart components stay thin and feature-specific.

4. Centralized state management — service-based signal stores or NgRx for complex cases. A **single source of truth** prevents state drift when 5+ components manipulate shared data.

— for design system + utilities. Teams consume versioned packages, not raw source — enables independent upgrades.

- **Lazy loading + @defer** — initial bundle stays small no matter how many features exist
- **OnPush + Signals** — change detection scales with app size
- **Virtual scrolling** for large lists — rendering stays O(visible rows), not O(total rows)
- **SSR + hydration** — fast TTFB and good SEO at scale
- **Caching (shareReplay, HTTP interceptor, service worker)** — reduce backend load
- **Strict mode + strict templates** — compiler catches bugs before they reach PRs
- **Automated testing** — unit + integration tests at feature boundaries
- **CI pipelines** — linting, budgets, bundle analysis, e2e smoke tests on every PR
- `ng update` — automated Angular version upgrades keep the stack modern
- **Monitoring** — Sentry/Datadog for runtime errors; Lighthouse budgets for performance regressions

On my largest project — 18 developers, 400+ screens — the three things that made scaling manageable were:

1. **Nx monorepo** with enforced module boundaries (no circular deps)
2. **Lazy-loaded feature libs** with their own data-access layer
3. **Performance budgets** in CI that failed any PR adding >20KB to initial bundle

Without these, the app would have drowned in bundle bloat and merge conflicts within 6 months.

- ☐ Feature-first folder structure
- ☐ Every feature is lazy-loaded
- ☐ `OnPush` by default on new components
- ☐ Design system in a shared library
- ☐ Bundle budgets in `angular.json`
- ☐ Automated tests on critical paths

- ☐ Observability (errors + performance metrics) in production
- *When should you move to an Nx monorepo?* — Once you have 2+ apps sharing code, or 10+ developers working on one codebase. Nx adds tooling overhead but pays off massively at scale.
- *What's the #1 anti-pattern that hurts scalability?* — **Circular feature imports** — Feature A imports Feature B, which imports Feature A. Kills build times, breaks lazy loading, and makes the code impossible to reason about.

QUICK REVISION CHEAT SHEET

Topic	Key Points to Remember
Compilation	JIT = browser/runtime (dev), AOT = build time/production (faster, smaller)
Components	Building blocks with TS + HTML + CSS, tree structure
Data Binding	Interpolation <code>{{ }}</code> , Property <code>[]</code> , Event <code>()</code> , Two-way <code>[]()</code>
Directives	Structural (<code>*ngIf</code> , <code>*ngFor</code>) + Attribute (<code>ngClass</code> , <code>ngStyle</code>)
Pipes	Transform display data, async pipe auto-unsubscribes, prefer pure pipes
Services	Singleton with <code>providedIn: 'root'</code> , share logic/data, keep components thin
DI	Angular auto-injects dependencies, hierarchical injectors, loosely coupled
Routing	RouterModule, Lazy Loading, Guards (CanActivate), Resolvers
Forms	Template-driven (simple) vs Reactive (complex), FormBuilder, Validators
HTTP	HttpClient returns Observables, Interceptors (auth token, error handling)
RxJS	<code>switchMap</code> (search), <code>debounceTime</code> , <code>takeUntil</code> (cleanup), <code>BehaviorSubject</code>
Lifecycle	<code>ngOnInit</code> (setup/API), <code>ngOnDestroy</code> (cleanup/unsubscribe)
Performance	Lazy load, OnPush, trackBy, virtual scroll, async pipe, avoid template functions
State	BehaviorSubject in services (lightweight), NgRx (complex), Signals (Angular 16+)
Modern Angular	Standalone components, Signals, new control flow (<code>@if</code> , <code>@for</code>), functional guards

INTERVIEW TIPS

1. **Always start with a direct answer** — then explain the why and how
2. **Connect every answer to your project** — "In my project, I used this for..."
3. **Don't bluff** — If you don't know, say "I haven't used it yet, but I understand the concept and can pick it up quickly"
4. **Use power keywords** — Interviewers listen for terms like "OnPush", "lazy loading", "switchMap", "BehaviorSubject", "tree-shakable", "standalone"
5. **Practice speaking out loud** — Read these answers aloud, not just in your head. Confidence comes from **practice, not memorization**
6. **Be ready to write code** — Interviewers may ask you to write interceptors, guards, reactive forms, services, or RxJS pipelines on the spot
7. **Show problem-solving mindset** — Don't just say what you used, explain **why** you chose that approach and what **alternatives** you considered

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